

RESEARCH ARTICLE

An SPH framework for drained and undrained loading over large deformations

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Abstract

We propose a new approach for performing drained and undrained loading of elastoplastic geomaterials over large deformations using smoothed particle hydrodynamics (SPH), a meshfree continuum particle method, combined with the modified Cam Clay (MCC) model of critical state soil mechanics. The numerical approach draws upon a novel one-particle two-phase penalty-method based formulation for handling undrained loading in saturated soils, which allows tracking of the buildup of pore-water pressures under combined shearing and compression. Large-scale parallelized simulations are employed to accommodate a significant number of degrees of freedom in a three-dimensional setting. After verification and benchmark testing, the SPH based formulation is used to analyze the propagation of reverse faults through fluid-saturated clay deposits and the rupture of strike-slip faults across earthen embankments. The computational methodology tests the robustness of the meshfree approach in situations where the soil tends to dilate on the ‘dry’ side of the critical state line and to compact on the ‘wet’ side, but cannot, because of the incompressibility constraint imposed by undrained loading. Our results extend the current understanding of fault rupture modeling and further demonstrate the potential of our framework together with the SPH method for large deformation analyses of complex problems in geotechnics.

KEYWORDS

embankment failure, fault rupture and propagation, penalty method, smoothed particle hydrodynamics, undrained loading

1 | INTRODUCTION

Because of the porous nature of geomaterials, geotechnical engineering design commonly involves accounting for fluid-saturated soils, fluid flow, and the subsequent pore water induced alteration of soil behavior and deformation. The stability of slopes, tunnels, foundations, excavations, or engineered earth structures can be susceptible to excess pore pressure buildup or seepage forces. An extreme design case occurs whenever the application of load to saturated soil is conducted under conditions inhibiting drainage, or alternatively, sufficiently rapidly such that pore pressures cannot begin to dissipate prior to the completion of load application. Engineering examples involving such undrained conditions, where both the soil can only deform in shear and not volumetrically, are numerous.

Traditionally, undrained analyses have been performed numerically with the finite element method (FEM)^{1,2} but for problems involving larger deformations and necessitating modeling post-failure behavior, the accuracy of the solution decreases due to mesh distortion and associated issues. Various different continuum particle-based or meshfree methods have been proposed in recent decades as alternatives to FEM, most notably the material point method (MPM),^{3–5} peridynamics,^{6–8} particle FEM (PFEM),^{9–11} element free Galerkin method (EFG),^{12–14} reproducing kernel particle method (RKPM),^{15–17} and smoothed particle hydrodynamics (SPH).^{18–20} As one of the best established of these methods, developed first by the work of Lucy²¹ and Gingold and Monaghan,²² SPH is one of the few truly meshfree methods, in that it requires no background mesh or knowledge of connectivity between the material points. In addition, since SPH is a fully collocational method, in contrast to FEM for example, SPH particles serve both as material points carrying field values, and as mathematical points where the equations of motion and constitutive models are solved. Since its early days centered on astrophysical applications and free surface flows, SPH has been applied to a variety of problems in geomechanics and geotechnics over the last decade involving the elastoplastic deformation of geomaterials. A subset of these applications worthy of mention include slope failures,^{23–26} debris flows,^{27–30} granular flows,^{31–34} and fault and fracture propagation.^{35–38}

To accurately model the strong coupling between pore fluid and the solid skeleton of soil or other geomaterials, a number of different formulations based on two distinct approaches have been proposed in SPH. In the multi-layer particle approach, each distinct phase is represented by its own set of particles, with the governing equations solved separately at each set of particles (phase). Alternatively, in the single-layer approach, one set of particles represents all the phases in the porous medium, and the Biot-Zienkiewicz equations using mixture theory are solved, and $\mathbf{u} - p$ and $\mathbf{u} - \mathbf{w} - p$ formulations where $\mathbf{u}$ is the solid skeleton displacement, p is the fluid pressure, and $\mathbf{w}$ is Darcy's velocity, have been recently proposed for both saturated and unsaturated soils.^{39–41} Nevertheless, these formulations in SPH can be computationally costly compared with coupled solid-fluid solvers in FEM since the SPH equations generally cannot be solved in conjunction, that is, monolithically, and projection methods consisting of a predictor-corrector schemes must be used to enforce incompressibility.³⁹

Due to the preponderance of undrained failure problems in geotechnics, many of which involve deformations beyond the scope which can be accurately handled in FEM, in this paper we propose a new single particle two-phase penalty-method based formulation for modeling solid-fluid coupled deformation in undrained soils using SPH. The proposed formulation has two main advantages over existing $\mathbf{u} - p$ and $\mathbf{u} - \mathbf{w} - p$ formulations: (1) it is easier to implement into any existing single phase SPH code, and (2) it has increased computational efficiency due to its simplicity. The parallel SPH code GEOSPH developed by the authors and built on the open-source framework PySPH⁴² is employed to implement the penalty formulation and perform our simulations. The parallel capabilities of our code allow for large-scale three-dimensional (millions of particles) simulations in reasonable computation time as shown in this paper. For more details regarding the code, its implementation, and past applications, the reader is referred to ref. [35, 36, 43, 44].

As an application of our formulation, we focus on the propagation of shallow lying faults through saturated soil under undrained conditions, particularly clay soils, as well as on quantifying damage and deformation to engineered earth structures overlying undrained soils. The rupture of faults in bedrock and their subsequent propagation through overlying saturated clay deposits, occurs on a time scale substantially smaller to that of any excess pore pressure dissipation that may have built up as a result of the fault rupture.⁴⁵ Therefore, any analysis of fault propagation through saturated clay performed under undrained conditions is valid as long as long-term soil consolidation or deformation effects are not of interest, which is generally true in fault rupture modeling as the emphasis is to determine the immediate post-rupture deformation of soil and earth structures which may have been damaged by the propagating fault, making this an ideal problem for our proposed framework. Since the risk posed to overlying structures and utilities by potentially active nearby shallow lying faults in bedrock is widely recognized especially in light of several recent earthquakes, for example, the Kocaeli and Chi-Chi earthquakes of 1999, a number of computational and analogue studies have been performed to try to quantify the trajectory taken by propagating faults through soil, and the deformation caused by fault rupture.^{35,45–52} In the case of reverse faults, few simulations have considered fault slips or offset beyond that required for initial surface rupture, in part due to the limitations of the traditional FEM, and even fewer studies have considered the presence of fluid saturation in the soil, especially in computational studies, despite the ubiquity of saturated soils near active fault zones.^{53–55} The soft soils considered in this study are modeled using the modified Cam Clay (MCC) constitutive model due to its ability to account for plastic volumetric compaction and the buildup of fluid pore pressures under compression. In doing so we develop a variation of the MCC closest point projection (CPP) return mapping algorithm amenable to the SPH method. The meshfree nature of SPH combined with the ability to account for pore pressure buildup in our proposed framework, helps us begin to address some of these previously mentioned shortcomings. To the best of our knowledge,

the combination of the proposed MCC implementation with an undrained penalty coupled formulation is novel within the SPH literature.

The sequence of presentation is as follows: in Section 2.1 of the paper, the governing and constitutive equations are presented, while their discretization using the SPH method and the structure of the proposed framework are detailed in Section 2.2. The adopted return mapping of the MCC model is described in Section 2.3 and the framework and its parallelized implementation is verified against a variety of benchmark tests possessing analytical solutions including triaxial tests under drained and undrained conditions in Section 3. With the framework having been successfully verified, it is next applied to study the propagation of reverse faults in homogeneous clay deposits in plane-strain in Section 4, as well as in full three-dimensional simulations modeling the rupture of strike-slip faults across earthen embankments in Section 5. Section 6 discusses our findings, and suggests areas for future work.

2 | GOVERNING EQUATIONS AND SPH FRAMEWORK

2.1 | Governing equations

The SPH method solves the mechanical initial boundary value problem (IBVP) on a domain $\mathcal{B}$ with boundary $\partial\mathcal{B}$ and natural and essential boundary conditions on $\partial\mathcal{B}_h \subset \partial\mathcal{B}$ and $\partial\mathcal{B}_v \subset \partial\mathcal{B}$ respectively, such that $\bar{\mathcal{B}} = \mathcal{B} \cup \partial\mathcal{B}$, $\partial\mathcal{B} = \partial\mathcal{B}_v \cup \partial\mathcal{B}_h$, and $\emptyset = \partial\mathcal{B}_v \cap \partial\mathcal{B}_h$. The strong form of the IBVP is posed as follows:

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho \dot{\mathbf{v}} & \forall \mathbf{x} \text{ in } \mathcal{B} \times t \\ \dot{\rho} = -\rho \nabla \cdot \mathbf{v} & \forall \mathbf{x} \text{ in } \mathcal{B} \times t \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{h} & \text{on } \partial\mathcal{B}_h \times t \\ \mathbf{v} = \bar{\mathbf{v}} & \text{on } \partial\mathcal{B}_v \times t, \end{cases} \quad (1)$$

with initial conditions $\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0$, $\mathbf{v}(\mathbf{x}, 0) = \mathbf{v}_0$, and $\dot{\mathbf{v}}(\mathbf{x}, 0) = \dot{\mathbf{v}}_0$, $\mathbf{h}(\mathbf{x}, 0) = \mathbf{h}_0$, $\rho(\mathbf{x}, 0) = \rho_0$ and $\boldsymbol{\sigma}(\mathbf{x}, 0) = \boldsymbol{\sigma}_0$ at time $t = 0$. Here, $\dot{\mathbf{v}}$ is the material time derivative of the velocity field, $\mathbf{b}$ is the body force (gravity) vector, $\mathbf{u}$ is the displacement field vector, $\mathbf{n}$ is the unit normal vector to boundary $\partial\mathcal{B}_h$, and $\boldsymbol{\sigma}$ is the total Cauchy stress tensor. The material density of the mixture is defined as $\rho = (1 - n)\rho_s + S_r n \rho_w$ where ρ_s is the density of the solid phase and ρ_w is the density of the pore fluid (water), n is the porosity, and S_r is the degree of saturation of the soil⁵⁶ ($0 \leq S_r \leq 1.0$ and $S_r = 1.0$ for fully saturated soil). Lastly, the vectors $\bar{\mathbf{v}}$ and $\mathbf{h}$ are the prescribed velocity and traction, respectively.

We note that in the IBVP the conservation of mass must be explicitly solved in conjunction with the conservation of linear momentum in the standard SPH formulation. Since the mass of any particle is kept constant throughout the simulations, the density of every particular particle is updated over time due to the deformation of the domain. A measure of deformation (rate) is the velocity gradient, defined as $\mathbf{l} = \partial\mathbf{v}/\partial\mathbf{x}$. The rate of deformation tensor $\mathbf{d}$ and spin tensor $\boldsymbol{\omega}$ are the symmetric and skew-symmetric parts of $\mathbf{l}$, that is,

$$\mathbf{d} = \frac{1}{2}(\mathbf{l} + \mathbf{l}^T), \quad \boldsymbol{\omega} = \frac{1}{2}(\mathbf{l} - \mathbf{l}^T). \quad (2)$$

The deformation rate tensor is assumed to be partitioned additively into elastic and plastic parts, $\mathbf{d} = \mathbf{d}^e + \mathbf{d}^p$.

The total Cauchy stress tensor $\boldsymbol{\sigma}$ for a fluid-saturated medium can be decomposed as follows,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - \alpha p^w \mathbf{1} \quad (3)$$

where $\boldsymbol{\sigma}'$ is the effective stress tensor, p^w is the pore fluid (assumed to be water) pressure, and $\mathbf{1}$ is the identity tensor. Note the Biot coefficient, α , is assumed to be one.⁵⁷ Under drained conditions, $p^w = 0$, and the total stress reduces to the effective stress, $\boldsymbol{\sigma} = \boldsymbol{\sigma}'$.

Under undrained conditions, we assume that the effective stress $\boldsymbol{\sigma}'_{n+1}$ is driven by the deformation rate $\mathbf{d}_{n+1}$ and can be evaluated using an elastoplastic constitutive model and standard return mapping algorithms. We further assume that pore pressure p^w can be expressed as a linear function of the volumetric strain, $\text{tr}(\boldsymbol{\epsilon})$, such that at time $t + 1$,

$$p^w_{n+1} = -K_w \text{tr}(\boldsymbol{\epsilon}_{n+1}), \quad (4)$$

where K_w is the bulk modulus of the fluid (water), and functions as the penalty parameter. We adopt the continuum mechanics sign convention where compression is negative, tension is positive, but pore pressure is positive under compression.

A hypo-elastoplastic model⁵⁸ defined by a direct relation between the Jaumann stress rate tensor in terms of the effective stress and the rate of deformation tensor is postulated as such,

$$\bar{\sigma}' = \mathbf{c}^{\text{ep}} : \mathbf{d}. \quad (5)$$

The Jaumann stress rate $\bar{\sigma}'$ is an objective stress rate that is invariant to rigid-body rotation, and is required given that the Cauchy stress rate is susceptible to changes under rotations in a fixed coordinate system. The Jaumann rate in terms of the effective stress is expressed as,

$$\bar{\sigma}' = \dot{\sigma}' - \omega \cdot \sigma' + \sigma' \cdot \omega. \quad (6)$$

In Equation (5), $\mathbf{c}^{\text{ep}}$ is the elastoplastic stress-strain response tensor. If the material response is elastic then $\mathbf{c}^{\text{ep}} = \mathbf{c}^{\text{e}}$, where for an isotropic material, $\mathbf{c}^{\text{e}}$ is written in terms of the elastic bulk and shear moduli K and μ , respectively,

$$\mathbf{c}^{\text{e}} = K \mathbf{1} \otimes \mathbf{1} + 2\mu \left(\mathbf{I} - \frac{1}{3} \mathbf{1} \otimes \mathbf{1} \right). \quad (7)$$

Here $\mathbf{I}$ is the fourth rank symmetric identity tensor with components $I_{ijkl} = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/2$ and $\mathbf{1}$ is the second-order identity tensor (Kronecker delta). Two different yield criteria are adopted in this study to simulate the elastoplastic behavior of granular fill-like materials and clays, the Drucker-Prager (DP) and MCC models. In the DP model, the elastic moduli are constant, whereas in the MCC model they are state variables which vary through time. We note that although widely used in the SPH modeling literature, the Jaumann rate may be susceptible to oscillations under simple shear.⁵⁹ In our simulations, however, no such oscillations were observed. Other objective stress rates may be employed, such as that based on the Lie derivative.⁶⁰ Before we further elaborate on the constitutive models mentioned, we will next provide an overview of the SPH method used to solve the IBVP and of our numerical framework.

2.2 | SPH discretization

The smoothed particle hydrodynamics (SPH) method is a continuum Lagrangian meshfree particle method. In SPH, the problem domain is discretized by a set of arbitrarily distributed particles that serve both as mathematical points to discretize and solve the governing equations, and as Lagrangian points that carry physical characteristics of the materials modeled, such as mass, mass density, stress, and strain. There is no predefined connectivity between particles and the underlying idea of the method is that interactions between particles are mediated through a weighting function or kernel, W , whose value for a given interaction depends on the distance between the two particles and a length scale called the smoothing length (h), which influences the accuracy and convergence characteristics of the method. In this paper, we use the Wendland C^2 kernel which does not suffer from the pairing instability seen in other commonly used kernels.⁶¹ The kernel is defined with respect to the deformed current configuration and is therefore an Eulerian kernel. An advantage of Eulerian kernels is that they are generally not susceptible to the “hourglassing” phenomenon which is known to affect the stability of the total Lagrangian SPH.⁶²

Given a field function $f(\mathbf{x}, t)$, where $\mathbf{x}$ is the position vector in three dimensions, and t is time, the approximation of the field function and of its first derivative with respect to $\mathbf{x}$ are given as

$$\langle f(\mathbf{x}) \rangle_i = \sum_{j=1}^N f(\mathbf{x}_j) W_{ij} V_j, \quad (8)$$

$$\langle \nabla f(\mathbf{x}) \rangle_i = \sum_{j=1}^N f(\mathbf{x}_j) \nabla W_{ij} V_j, \quad (9)$$

where $\langle \cdot \rangle$ signifies approximation, the subscript j represents the neighboring particles of particle i , located at $\mathbf{x} = \mathbf{x}_i$, $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)_i$ is the vector of partial derivatives with respect to the three spatial coordinates, evaluated at the position of particle i , $W_{ij} = W(\mathbf{x}_i - \mathbf{x}_j, h)$, and $V_j = m_j/\rho_j$ is the volume occupied by particle j , with m_j and ρ_j the mass and mass density of the particle, respectively. Note that the summation is performed over all neighboring particles of particle i , including itself.

The smoothing length, h , also defines the support of the kernel and the radius of interaction of particle i . Only particles within a radius $k_h h$ of particle i are considered neighbors (j) and included in the summations in Equations (8) and (9). The coefficient $k_h = 2.0$ is usually chosen in most SPH applications.³¹ In effect, the smoothing length is a numerical characteristic length scale intrinsic to SPH, which provides the method with nonlocal properties.

More complete overviews of the SPH method, its formulations and derivations, can be found in ref. 43, 61, 63–65. Based on the basic operators in Equations (8) and (9), the most commonly adopted discretized forms of the balance equations in the IBVP are as follows:

$$\langle \dot{\mathbf{v}} \rangle_i = \sum_{j=1}^N m_j \left(\frac{\boldsymbol{\sigma}_i + \boldsymbol{\sigma}_j}{\rho_i \rho_j} \right) \cdot \nabla W_{ij} + \mathbf{b}_i, \quad (10)$$

$$\langle \dot{\rho} \rangle_i = \sum_{j=1}^N m_j (\mathbf{v}_j - \mathbf{v}_i) \cdot \nabla W_{ij}. \quad (11)$$

In practice, due to its explicit and dynamic nature, SPH particles may oscillate and get too close to one another, which requires some numerical damping and viscosity to mitigate the problem. In this work, we utilize the well-know Monaghan artificial viscosity⁶⁶ and a damping force into the balance of linear momentum, Equation (10). Hence, the balance of linear momentum becomes

$$\langle \dot{\mathbf{v}} \rangle_i = \sum_{j=1}^N m_j \left(\frac{\boldsymbol{\sigma}_i + \boldsymbol{\sigma}_j}{\rho_i \rho_j} + \boldsymbol{\Pi}_{ij} \right) \cdot \nabla W_{ij} + \mathbf{b}_i - c_d \mathbf{v}_i, \quad (12)$$

where $\boldsymbol{\Pi}_{ij}$ is the artificial viscosity of Monaghan, with coefficients $0.1 \leq \alpha_\pi \leq 0.8$ and $0 \leq \beta_\pi \leq 0.4$, and c_d is the damping coefficient given by $c_d = \xi \sqrt{E/(\rho h^2)}$, with E the elastic modulus of the material, and $0 \leq \xi \leq 0.05$.

When the soil is fully saturated, we assume an undrained behavior of the sediment in our simulations. Hence, during deformation, excess pore-water pressures may arise, which are calculated in rate form from Equation (4) by substituting the volumetric strain for the volumetric rate of deformation, $\text{tr}(\mathbf{d})$:

$$\langle \dot{p}^w \rangle_i = -K_w \sum_{j=1}^N V_j (\mathbf{v}_j - \mathbf{v}_i) \cdot \nabla W_{ij}, \quad (13)$$

where the summation term is the SPH approximation to the velocity divergence, $\nabla \cdot \mathbf{v} = \text{tr}(\mathbf{d})$.

Since the volumetric deformation experienced by any given particle can be positive (volume increase) or negative (volume decrease), pore pressures calculated using Equation (13) may result in negative (suction) or positive (compression) numbers. To avoid numerical instabilities and since the penalty formulation is not intended to determine pore pressures associated with suction, we set a cutoff for pore pressures, such that if any particle's final pore pressure becomes negative, it is immediately set to zero. Due to the presence of pore-water pressure, the total Cauchy stress tensor used in solving Equation (12) is then calculated as:

$$\boldsymbol{\sigma}_i = \boldsymbol{\sigma}'_i - p_i^w \mathbf{1}, \quad (14)$$

where the effective Cauchy stress tensor $\boldsymbol{\sigma}'$ comes from the constitutive model update.

Similarly to what was presented for the balance equations, the velocity gradient tensor used to calculate the deformation rate tensor is discretized as

$$\langle \mathbf{l} \rangle_i = \sum_{j=1}^N V_j (\mathbf{v}_j - \mathbf{v}_i) \otimes \nabla W_{ij}, \quad (15)$$

with $\otimes$ the outer product operator.

In order to improve the consistency of the approximations for Equations (11), (13), and (15), a modified version of the kernel gradient is applied, given by

$$\tilde{\nabla} W_{ij} = \mathbf{L}_i^{-1} \cdot \nabla W_{ij}, \quad (16)$$

with

$$\mathbf{L}_i^{-1} = \left[\sum_{j=1}^N V_j (\mathbf{x}_i - \mathbf{x}_j) \otimes \nabla W_{ij} \right]^{-1}. \quad (17)$$

The corrected kernel gradient is then substituted in place of the regular kernel gradient $\tilde{\nabla} W_{ij} \rightarrow \nabla W_{ij}$ in Equations (11), (13), and (15). The balance of momentum equation (Equation 10) is not corrected as the approximation errors induced whenever the kernel gradient is not corrected are required to guarantee the conservation of both linear and angular momenta.⁶¹

When computing the balance of linear momentum, the balance of mass, and the velocity gradient, the summations over kernels for domain particles near the boundaries are truncated for lack of particles filling out the kernel domain. This truncation leads to loss of consistency and accuracy in the approximations, so boundary particles are used to complete the kernel while enforcing essential boundary conditions.

In this work, we use the so-called dummy boundary conditions (also known as fixed boundary), where three to four layers of particles are placed where solid walls would be physically present. These particles either move with prescribed velocities or are fixed in space, and their mass densities and masses are kept constant throughout the simulation. The only parameters that are evolved over time are the effective stress tensor and pore-water pressure. We determine the effective stress tensors of the boundary particles based on stresses from the domain using the formulation presented in ref. 33 to enforce no-slip or free-slip conditions on the solid boundaries.

Pore-water pressures in boundary particles are calculated by simply extrapolating the pore pressures of neighboring soil particles onto boundary particles using Adami et al.'s formulation,⁶⁷ as follows:

$$p_b^w = \sum_{s=1}^{N_b} V_s (p_s^w + z g \rho_w) \tilde{W}_{bs}, \quad (18)$$

where subscripts b and s refer to boundary and domain (soil) particles, respectively, N_b is the number of domain neighboring particles of boundary particle b , z is the distance in the direction of gravity between boundary and domain particles, and g is the gravitational acceleration (-9.81 m/s^2 in our simulations). The second term within the parenthesis helps with the creation of a smooth pressure field under hydrostatic conditions. The term $\tilde{W}$ is the so-called normalized kernel, calculated as

$$\tilde{W}_{bs} = \frac{W_{bs}}{\sum_{s=1}^{N_b} V_s W_{bs}}. \quad (19)$$

The simulation is marshalled dynamically and explicitly in time, where the known information at time $t = t_n$ corresponding to step n is used to determine the updated variables at time $t = t_{n+1}$ at step $n + 1$. We use a semi-implicit time integration scheme for its best conservation characteristics.⁶⁵ In this scheme, the particle velocity is updated first and then its position and density, as follows:

$$\mathbf{v}_{i,n+1} = \mathbf{v}_{i,n} + \dot{\mathbf{v}}_{i,n} \Delta t, \quad (20)$$

$$\mathbf{x}_{i,n+1} = \mathbf{x}_{i,n} + \mathbf{v}_{i,n+1} \Delta t, \quad (21)$$

$$\rho_{i,n+1} = \rho_{i,n} + \dot{\rho}_{i,n} \Delta t, \quad (22)$$

where $\Delta t = t_{n+1} - t_n$ is the time step which has to obey the CFL condition to ensure stability of the simulation. The time step is calculated as follows:

$$\Delta t = C_0 \frac{h}{c_s}, \quad (23)$$

where C_0 is the CFL coefficient, taken as 0.1, and $c_s = \max \left\{ \sqrt{E_0/\rho_0}, \sqrt{K_w/\rho_0} \right\}$ is the numerical sound speed of the material, with E_0 , ρ_0 , and K_w the reference elastic modulus and mass density of the material, and the numerical bulk modulus of water, respectively.

We also need to update the effective stress tensor and the pore pressure of soil particles at every time step, which is accomplished using the following equations:

$$\sigma'_{n+1} = \sigma'_n + \dot{\sigma}'_{n+1} \Delta t, \quad (24)$$

$$p_{n+1}^w = p_n^w + \dot{p}_n^w \Delta t. \quad (25)$$

The flow-chart in Figure 1 provides a high-level implementation of the proposed formulation. In what follows, we will provide details on the MCC model used in our work.

2.3 | Modified Cam Clay (MCC) model

The modified Cam Clay (MCC) model is well suited for modeling normally consolidated or overconsolidated soft clays.⁶⁸ In the model, strain hardening and softening are accounted for in the basic formulation without any need for modification or improvement. By introducing the concept of the critical state line (CSL), the model discriminates between yielding of the material and ultimate collapse. Through the nonlinear dependence of the void ratio on the mean effective stress and making use of the concept of the state boundary surface, a yield surface can be defined in $e - p' - q$ space, where e is the void ratio, $p' = \text{tr}[\sigma']/3$ is the mean effective stress, and $q = \sqrt{3/2} \|\mathbf{s}\|$ is the von Mises (deviatoric) stress, with σ' the effective Cauchy stress tensor, and $\mathbf{s} = \sigma' - p'\mathbf{1}$ the deviatoric stress tensor.

In $p' - q$ space the yield function for the MCC model is an ellipse given by

$$\mathcal{F} = \frac{q^2}{M^2} + p'(p' - p_c) \leq 0, \quad (26)$$

where the preconsolidation pressure $p_c < 0$, determines the diameter of the ellipsoid in the p' -axis.

At critical state, the material can be sheared without any volume change, and the locus of points satisfying this condition is given by the CSL, taking the form $q = -Mp'$, where its slope, M , is defined as,

$$M = \frac{6 \sin \varphi_{cs}}{3 - \sin \varphi_{cs}}, \quad (27)$$

in which φ_{cs} is the critical state internal friction angle of the material. The hardening law is given in rate form as

$$\dot{p}_c = -v \text{tr}[\mathbf{d}^p] p_c, \quad v = \frac{1+e}{\lambda - \kappa}, \quad (28)$$

where λ is the slope of the virgin isotropic consolidation line, and κ is the slope of the swelling line in $\log(p') - e$ space. Note e and v are assumed to be constant over a given time step, Δt . Likewise the elastic moduli,

$$K = \frac{(1+e)}{\kappa} p', \quad \mu = \frac{3K(1-2\nu)}{2(1+\nu)} \quad (29)$$

are treated as constant over Δt , and the Poisson's ratio ν is considered temporally invariant. Alternatively, the two elastic moduli may be derived from a theory based on hyperelasticity.⁶⁹

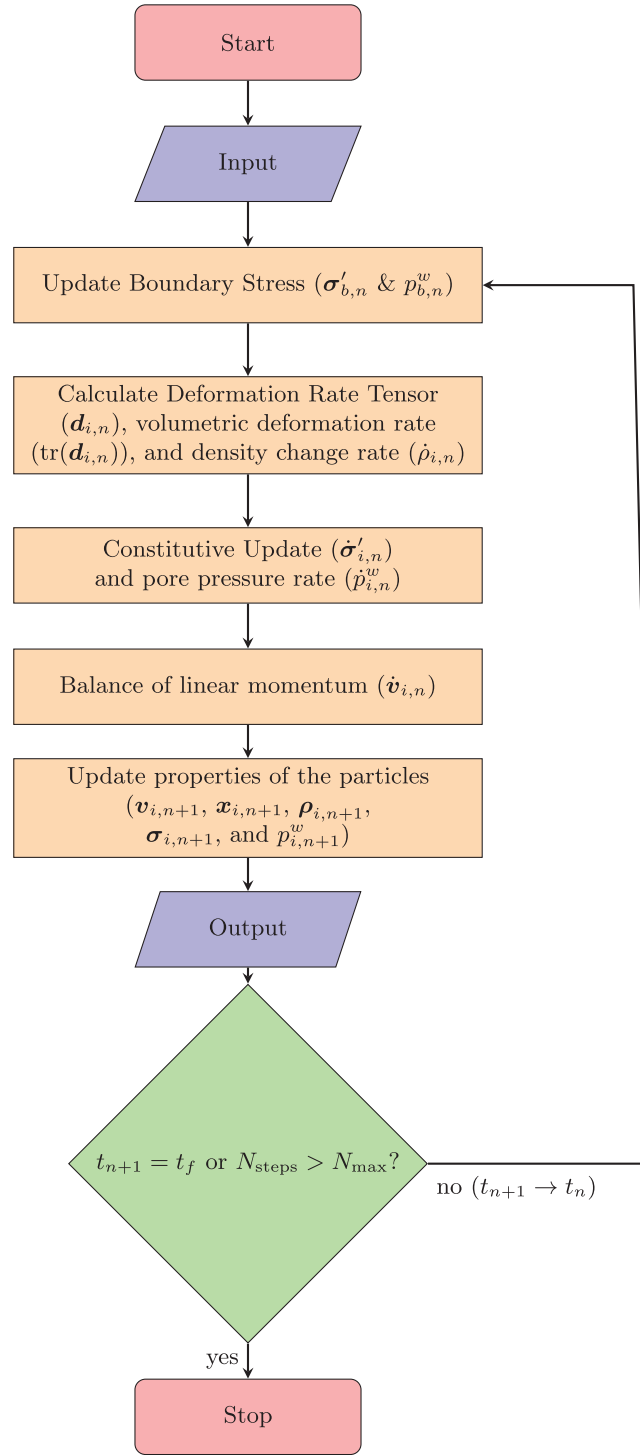


FIGURE 1 Flowchart of the SPH implementation of our proposed drained and undrained formulation.

An associative flow rule is employed such that,

$$\mathbf{d}^p = \dot{\phi} \frac{\partial \mathcal{F}}{\partial \boldsymbol{\sigma}'} = \frac{1}{3} (2p' - p_c) \mathbf{1} + \sqrt{\frac{3}{2}} \frac{2q}{M^2} \hat{\mathbf{n}}, \quad (30)$$

where ϕ is the plastic multiplier and $\hat{\mathbf{n}} = \mathbf{s}/\|\mathbf{s}\|$. Integrating the hardening law and flow rule implicitly and noting that $\text{tr}[\Delta \mathbf{d}_{n+1}^p] = \Delta \phi \text{tr}[\partial \mathcal{F}/\partial \boldsymbol{\sigma}'] = \Delta \phi (\partial \mathcal{F}/\partial p')$,

$$(p_c)_{n+1} = \frac{(p_c)_n}{1 + v \text{tr}[\Delta \mathbf{d}_{n+1}^p]} = \frac{(p_c)_n}{1 + v \Delta \phi (2p' - p_c)}. \quad (31)$$

A variation of the closest point projection (CPP) returning mapping as proposed by Borja and Lee⁷⁰ is utilized. Making use of the return mapping tensor equation,

$$\boldsymbol{\sigma}'_{n+1} = \boldsymbol{\sigma}'_{n+1}^{\text{tr}} - \mathbf{c}^e : \Delta \mathbf{d}^p \quad (32)$$

and an implicit time integration of the flow rule (Equation 30) permits the derivation of return mapping equations for the mean effective stress p' and deviatoric stress q ,

$$p' = \frac{p'_{n+1}^{\text{tr}} + K \Delta \phi p_c}{1 + 2K \Delta \phi}, \quad (33)$$

$$q = \frac{q_{n+1}^{\text{tr}}}{1 + 6\mu \frac{\Delta \phi}{M^2}}. \quad (34)$$

The consistency parameter $\Delta \phi$ is determined by imposing the consistency condition $\mathcal{F}(\Delta \phi) = q^2/M^2 + p'(p' - p_c) = 0$, and performing a local Newton iteration,

$$\Delta \phi^{k+1} = \Delta \phi^k - \frac{\mathcal{F}^k}{(\mathcal{F}')^k}, \quad (35)$$

with $\mathcal{F}' = \partial \mathcal{F}/\partial \Delta \phi$ the partial derivative of the yield function with respect to the consistency parameter. To obtain the root $\Delta \phi^k$ at the k th iteration, which satisfies the consistency condition, an initial estimate of $\Delta \phi^0 = 0$ is used, while simultaneously updating Equations (31), (33), and (34) from initial guesses of $p'_{n+1}^0 = p'^{\text{tr}}$, $q_{n+1}^0 = q^{\text{tr}}$, and $(p_c)_{n+1}^0 = (p_c)_n$, to obtain p'_{n+1}^k , q_{n+1}^k , and $(p_c)_{n+1}^k$. We note that integrating the preconsolidation pressure p_c approximately in Equation (31), decouples Equations (33) and (34), which is not the case in the original return mapping of Borja and Lee⁷⁰ (hereafter denoted as the exact CPP algorithm), and eliminates the need for an additional Newton iteration loop compared to the original exact algorithm, at an essentially negligible accuracy cost as will be subsequently shown. The entirety of the return mapping algorithm is summarized in Algorithm A1 in the Appendix section, presented alongside the partial derivatives of the yield function and the update equation for $(p_c)_{n+1}^k$ which all differ from the original exact CPP algorithm.

With $\Delta \phi$, p'_{n+1} , q_{n+1} , $(p_c)_{n+1}$, as well as K_{n+1} , μ_{n+1} , and v_{n+1} all determined at the conclusion of the Newton loop, and $\boldsymbol{\sigma}'_{n+1}^{\text{tr}}$ and $\mathbf{d}_{n+1}$ already known, $\boldsymbol{\sigma}'_{n+1}$ is found by first integrating Equation (30) as

$$\mathbf{d}_{n+1}^p = \frac{\Delta \phi}{\Delta t} \left(\frac{\partial \mathcal{F}}{\partial \boldsymbol{\sigma}'} \right)_{n+1} = \frac{\Delta \phi}{\Delta t} \left(\frac{1}{3} (2p'_{n+1} - (p_c)_{n+1}) \mathbf{1} + \sqrt{6} \left(\frac{q_{n+1}}{M^2} \right) \frac{\mathbf{s}_{n+1}^{\text{tr}}}{\|\mathbf{s}_{n+1}^{\text{tr}}\|} \right), \quad (36)$$

and then evaluating,

$$\boldsymbol{\sigma}'_{n+1} = \mathbf{c}_{n+1}^e : (\mathbf{d}_{n+1} - \mathbf{d}_{n+1}^p). \quad (37)$$

Time integrating the Cauchy stress rate gives,

$$\boldsymbol{\sigma}'_{n+1} = \boldsymbol{\sigma}'_n + \Delta t \boldsymbol{\sigma}'_{n+1} + \mathbf{R} \cdot \boldsymbol{\sigma}'_n - \boldsymbol{\sigma}'_n \cdot \mathbf{R}, \quad \mathbf{R} = \Delta t \boldsymbol{\omega}_{n+1} \quad (38)$$

The Jaumann stress rate $\boldsymbol{\sigma}'$ and the rotation tensor $\mathbf{R}$ are taken at time $n + 1$ since the velocity rate tensor ($\mathbf{l}_{n+1}$) and hence $\mathbf{d}_{n+1}$ and $\boldsymbol{\omega}_{n+1}$ are updated before the stress. The explicit computation of the rotation tensor $\mathbf{R}$ is contingent upon the material undergoing small rotation within the time increment. For small enough time steps, this requirement is generally fulfilled. Otherwise an objective algorithm for integrating rate constitutive equations must be used.⁷¹

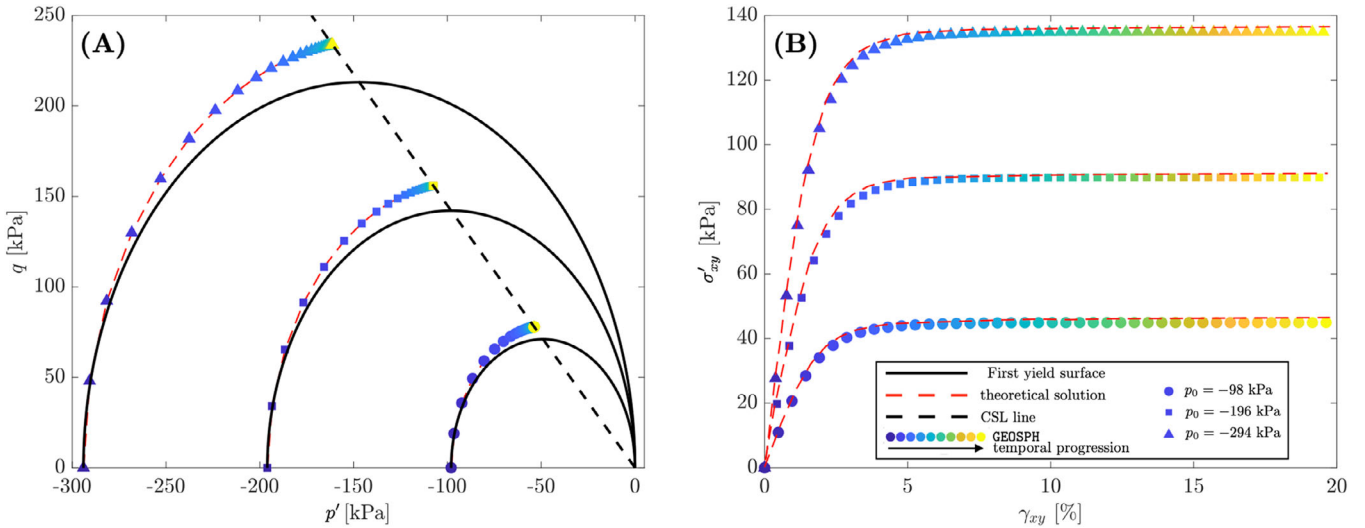


FIGURE 2 Comparison of the performance of the approximate CPP algorithm using GEOSPH against the theoretical solution of simple shear tests of normally consolidated soils, that is, $p'_0 = (p_c)_0$. In the first test $p'_0 = (p_c)_0 = -98$ kPa, in the second $p'_0 = (p_c)_0 = -196$ kPa, and in the third $p'_0 = (p_c)_0 = -294$ kPa.

3 | FORMULATION VERIFICATION

3.1 | Simple shear tests

To ensure that the SPH implementation can correctly simulate the constitutive response seen in the MCC model, a series of simple shear tests for both normally consolidated and overconsolidated soils are performed and compared against theoretical solutions. Simple shear tests are considered the preferred method to validate a constitutive model in SPH given standard element tests often performed in FEM analyses are not available in SPH.⁶¹ The test setup, dimensions, and material parameters are taken from Nonoyama et al.²³ Two subsets of the results are presented for brevity. In the first, shown in Figure 2, the SPH solutions using the approximate CPP algorithm are compared to theoretical solutions of simple shear tests for normally consolidated soils with different initial values of $p'_0 = (p_c)_0$. Here the SPH solution is shown to closely match the theoretical one. In Figure B1 (shown in Appendix B), the results of the approximate CPP algorithm are juxtaposed with those of the original exact CPP algorithm, for overconsolidated soil simple shear tests, rendering almost identical results with negligible error, at a slightly reduced computational cost, and without the convergence issues arising in the exact CPP algorithm due to its nested Newton loops under more complex boundary value problems in SPH.

3.2 | Self-weight undrained loading tests

A first and simple way to test the undrained penalty formulation is to perform the stress initialization of saturated one-dimensional elastic soil columns under the action of gravity (self-weight). The undrained nature of the loading path renders the effective vertical stress and pore-water pressure distributions distinct from those under drained conditions (geostatic/hydrostatic states), and analytical solutions are available to assess the results. The analytical solutions for the pore pressure and effective vertical stress over depth are, respectively⁷²:

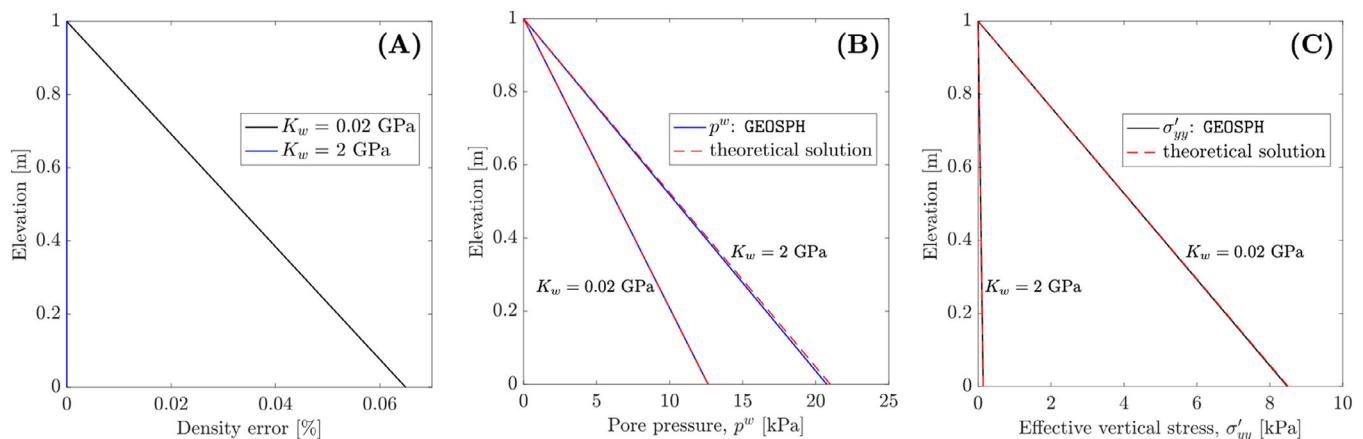
$$p_0^w(z) = \frac{K_w \rho g (H - z)}{K + \frac{4}{3} \mu + K_w}, \quad (39)$$

$$\sigma'_{v,0}(z) = \frac{\left(K + \frac{4}{3} \mu\right) \rho g (H - z)}{K + \frac{4}{3} \mu + K_w}, \quad (40)$$

where H is the height of the soil column and the coordinate z is measured from the bottom of the column upwards. In the simulation setup, the pore-water pressure of the top layer of soil particles is set to zero and fixed at that value throughout

TABLE 1 Self-weight undrained loading simulation parameters.

Parameter	Value
Time step, Δt [s]	1e-5 and 1e-6
Initial interparticle distance, Δ [m]	0.01
Smoothing length factor, k_h	1.5
Artificial viscosity parameters, α_π and β_π	0.8 and 0.0
Artificial stress parameter, ϵ	0.0
Damping coefficient, ξ	0.03
Soil column height, H [m]	1.0
Solid phase density, ρ_s [kg/m ³]	1850
Porosity, n	0.3
Mixture density, ρ [kg/m ³]	2155
Elastic modulus, E [Pa]	1e7
Poisson's ratio, ν	0.3
Bulk modulus of water, K_w [GPa]	0.02 and 2

**FIGURE 3** Self-weight undrained loading test results. (A) Density error, (B) pore-water pressure distribution, and (C) effective vertical stress.

the simulation. Moreover, the bottom of the soil column is fixed (no-slip boundary condition, while the sides of column are free to move in the vertical direction (free-slip or periodic). The simulation parameters are provided in Table 1.

Two simulations were performed following the work of Chen et al.,⁷² and two values of water bulk modulus were chosen, 0.02 and 2 GPa. The latter is close to the physical value of the bulk modulus for water. These values were chosen to assess the sensitivity of the proposed formulation to that parameter. Figure 3 shows the results of the density error, and the distribution of effective vertical stress and pore pressure for both bulk moduli. The density is a reliable proxy to validate the formulation in abiding to the assumption of quasi-incompressibility of the water phase. Quasi-incompressibility in this case is manifested through negligible changes in the mixture mass density (<1%).

Note that the formulation proposed is fairly insensitive to the chosen value of water bulk modulus, with results using $K_w = 0.02$ GPa yielding as good results as those using the actual bulk modulus of water, and both in extremely good agreement with the theoretical solutions. Additionally, as can be seen in panel (A) of Figure 3, quasi-incompressibility is strongly enforced with variations of the density of less than a 10th of a percent.

3.3 | Triaxial compression tests

As a final step in the verification process of the MCC constitutive model and of the penalty formulation and their implementation in GEOSPH, a series of triaxial tests are performed under drained and undrained conditions. Triaxial tests have

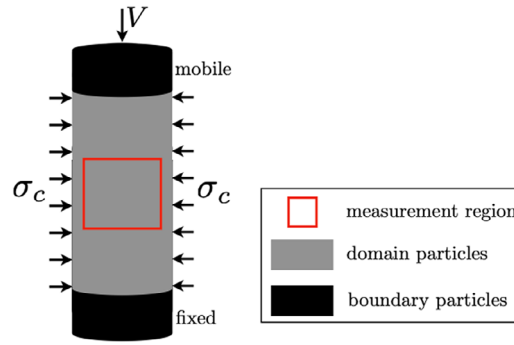


FIGURE 4 Schematic showing the setup of the triaxial simulations. The soil cylinder is discretized into domain particles. The confining stress, σ_c , is applied on the lateral boundaries of the cylinder, while a velocity V is prescribed to the boundary particles comprising the top cap of the soil cylinder.

been conducted using the SPH method by a number of authors, primarily using the DP or Mohr-Coulomb models, and under drained conditions.^{73,74} The work of Morikawa and Asai³⁹ successfully simulated drained and undrained triaxial tests using the MCC model, but employed a $\mathbf{u} - \mathbf{w} - p$ formulation, which can be quite computationally costly if only the end-case of undrained behavior is desired.

The main challenge in modeling triaxial tests with SPH lies in correctly enforcing confinement along the lateral boundaries of the cylinder. Traditionally, confining stress was applied in SPH by a laborious process involving determining the particles composing the free surfaces, then the normal vectors for each particle, and lastly, computing the surface area at each particle where the confining stress should be applied.⁷³ When dealing with large deformations such as in triaxial tests, it becomes increasingly difficult to correctly determine the normals, as well as the required surface area, leading to significant error. Alternatively, enveloping the domain particles in boundary or ghost type particles has the disadvantage that such particles may not be sufficiently flexible to accommodate the specimen bulging typical of triaxial tests, and such a procedure may also complicate applying the exact desired magnitude of confining stress.

As an alternative, in our triaxial simulations we employ the flexible confined boundary conditions proposed by Zhao et al.,⁷⁴ where a confining stress σ_c is applied directly to the conservation of momentum equation (Equation 12), to all the domain particles in the simulation,

$$\langle \dot{\mathbf{v}} \rangle_i = \sum_{j=1}^N m_j \left(\frac{\sigma_i + \sigma_j}{\rho_i \rho_j} + \Pi_{ij} \right) \cdot \nabla W_{ij} + \sum_{j=1}^N m_j \left(\frac{(\sigma_c)_i + (\sigma_c)_j}{\rho_i \rho_j} \right) \nabla W_{ij} + \mathbf{b}_i - c_d \mathbf{v}_i. \quad (41)$$

Because of the symmetry and compactness of the SPH kernels, and because they are truncated along the free surfaces, it can be shown using the divergence theorem that σ_c is in effect only applied on the free surface and vanishes by default at all other particles within the domain.⁷⁴ Therefore, the confining pressure can be applied without having to identify particles along the free-surfaces or their normals.

The soil cylinder used in the simulations is 0.15 m high and has a diameter of 0.05 m. It is discretized using 53,175 particles with an initial interparticle distance of 0.002 m. Boundary particles are placed at the top and bottom faces of the cylinder, with the bottom boundary particles fixed, and the top boundary particles subjected to a constant velocity $V = 0.01$ m/s in the vertical direction to compress the soil sample. The discretized geometry of the soil cylinder and the relevant boundary conditions are shown in Figure 4. Stress and deformation data is measured as an average taken from all the particles within a cube (denoted the measurement region) with a side length of 0.03 m located at the center of the specimen to ensure that the stress state of the particles matches a triaxial state and to smooth out and mitigate the effects of any particle disorder occurring within the sample. In total, six simulations are performed, three under drained (TD) and another three under undrained conditions (TU), with each consisting of a lightly overconsolidated soil test (L) a moderately overconsolidated test (N) and a normally consolidated test (M). A summary of the simulation parameters is provided in Table 2.

Figures 5, B2, and B3 (latter two displayed in Appendix B) show the loading path in the measurement region of the cylindrical specimen in $p' - q$ (Panel A) and $\log(p') - e$ (Panel B) space for the lightly overconsolidated, normally consolidated, and moderately overconsolidated tests under drained conditions. Likewise, panels (A) and (B) in Figures 6,

TABLE 2 Triaxial test simulation parameters.

Parameter	Value
Time step, Δt [s]	4.84e-6 and 7.7e-7
Initial interparticle distance, Δ [m]	0.002
Smoothing length factor, k_h	1.5
Artificial viscosity parameters, α_π and β_π	0.2 and 0.4
Artificial stress parameter, ϵ	0.2
Damping coefficient, ξ	0.05 (for undrained -U simulations)
Soil cylinder diameter, D [m]	0.06
Soil cylinder height, H [m]	0.15
Compaction velocity, V [m/s]	0.01
Initial density, ρ [kg/m ³]	3000
Poisson's ratio, ν	0.3
Critical state line slope, M	1.185
Initial void ratio, e	2.0
Virgin isotropic consolidation index, λ	0.2
Swelling index, κ	0.05
Initial preconsolidation pressure, $(p_c)_0$ [kPa]	-200
Initial mean effective stress (confining pressure), $p'_0 = \sigma_c$ [kPa]	-150 (-L), -200 (-N) and -30 (-M)
Bulk modulus of water, K_w [GPa]	0.2 (for undrained -U simulations)

Note: L stands for lightly overconsolidated, N for normally consolidated, and M stands for moderately overconsolidated clays.

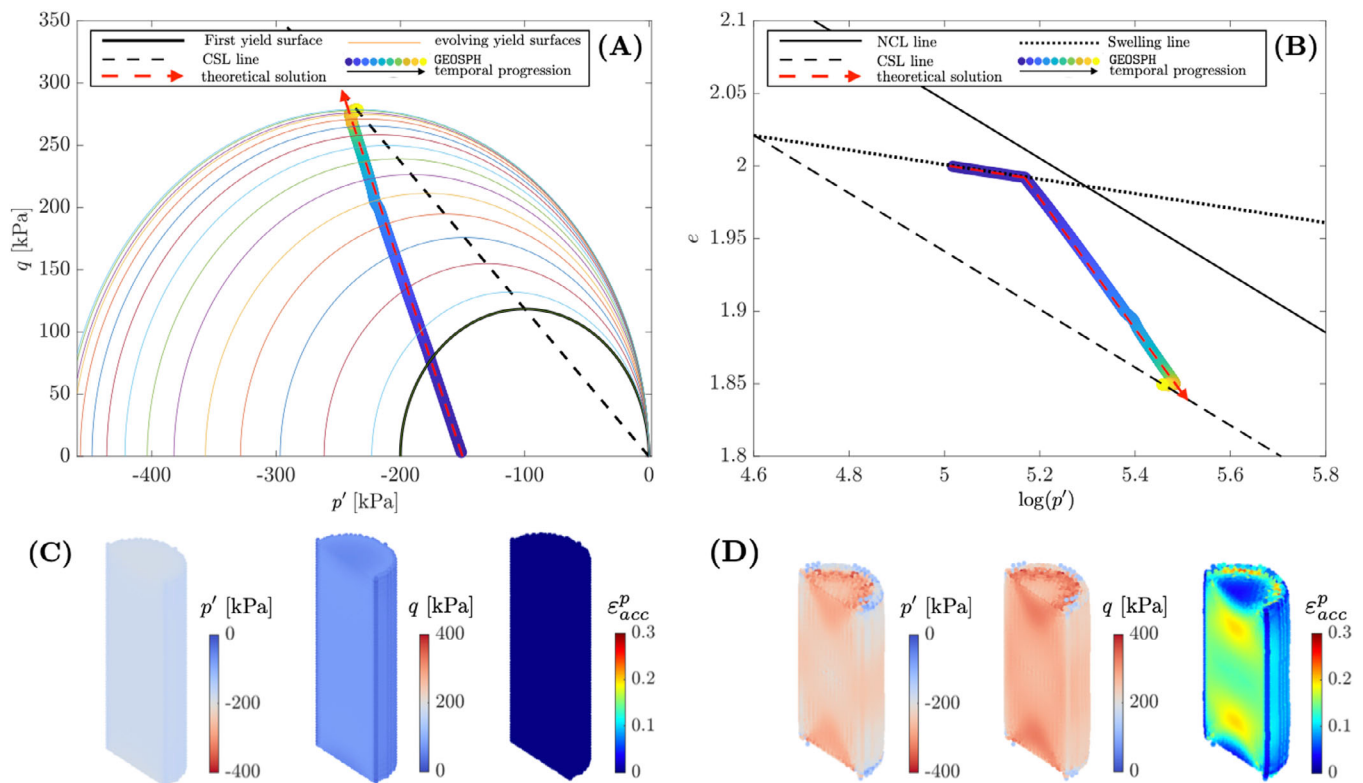


FIGURE 5 Results of drained triaxial test simulation using GEOSPH for a lightly overconsolidated soil (TD-L). The stress history in $p' - q$ space is shown in (A) whereas the trajectory in $\log(p') - e$ space is displayed in (B), and both are compared against the theoretical solutions as outlined by Wood.⁷⁵ The blue to yellow color scheme depicts the temporal evolution of the load path computed with SPH. Snapshots of the the mean effective stress p' , von Mises stress q and accumulated plastic strain ϵ^p_{acc} along the soil cylinder cut in half, are displayed at the onset of yielding prior to any plastic strain accumulation (C) and at the CSL locus (D).

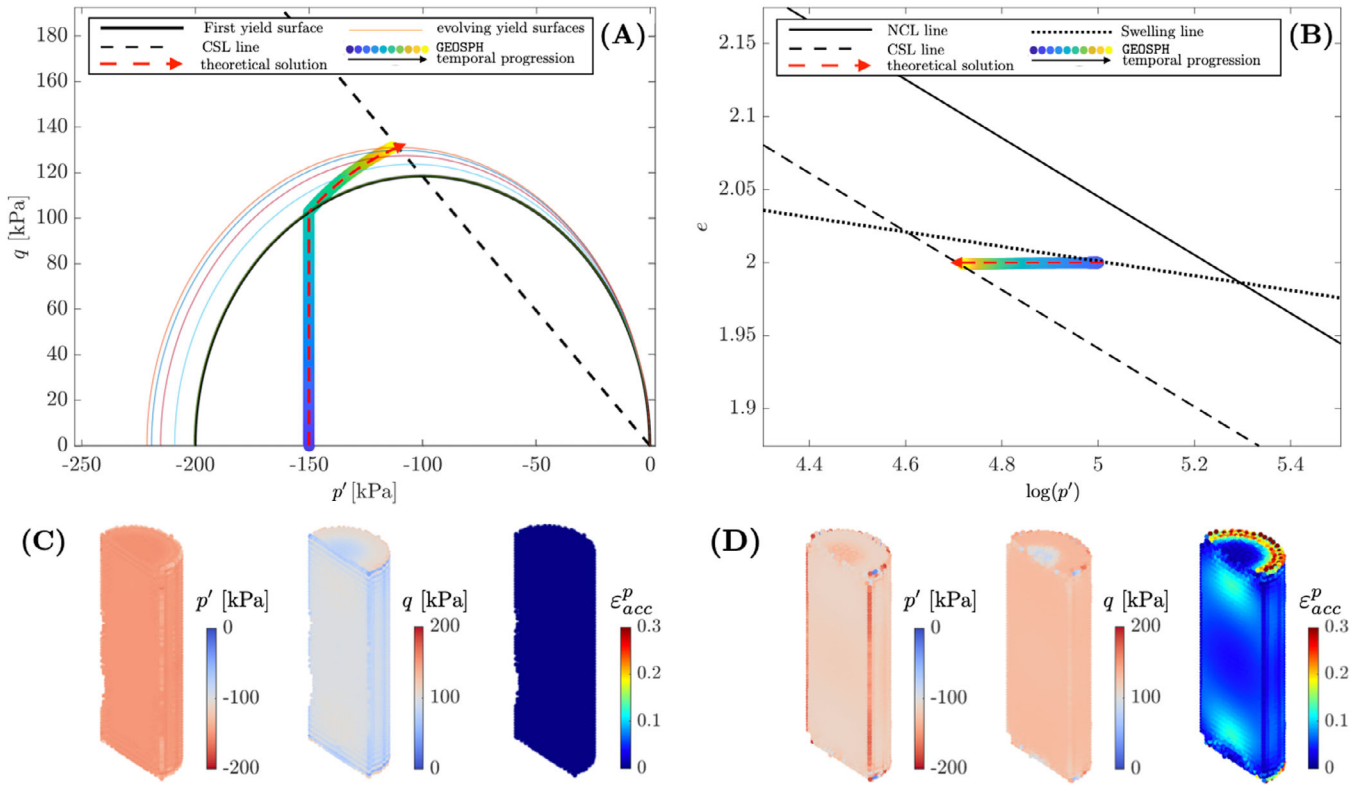


FIGURE 6 Same as Figure 5 but for the undrained lightly overconsolidated soil (TU-L) simulation.

B4, and B5 (latter two displayed in Appendix B) show the equivalent results for the undrained triaxial tests. For the drained tests the loading path carefully follows that of the theoretical solution, with the characteristic 3:1 slope, until the stress reaches the CSL. Conversely, in $\log(p') - e$ space, the numerical solutions follow the theoretical ones closely as well, initially adhering to the swelling line prior to yielding and then trending towards the CSL in the drained tests. For the undrained simulations, the SPH stress paths also closely match the theoretical loading paths in $p' - q$ space as well as in $\log(p') - e$ space where they exhibit incompressible behavior.

The evolution of the pore-water pressure p^w at the measurement region in the undrained tests is also compared against the theoretical solution in Figure 7, showing good agreement with the theory. As expected, the pore pressure accommodates the stresses not accounted for in the mean effective stress p' , and for the lightly overconsolidated (TU-L) and normally consolidated (TU-N) soils, asymptotes to a peak value as the stress reaches the CSL. In the former, a small kink is observed at the point of yielding, following theoretical expectations. For the moderately overconsolidated (TU-M) soil, yielding results in the shrinking of the yield locus and the difference between total and effective loading paths decreases, asymptotically reducing the pore pressure. Similar good agreement between the theoretical von Mises stress and that obtained from the SPH triaxial simulations is displayed in Figure B6 in Appendix B, with the stress converging to the ultimate theoretical von Mises stress q_{cs} at critical state conditions, as the CSL line serves as a stress attractor.

3.4 | High-performance computing (HPC) implementation

Since the algorithmic structure of SPH presents ample opportunities for parallel execution,^{76,77} parallel computing is a natural solution to improve computational performance, especially for three-dimensional simulations. Utilizing the parallel capability of PySPH,⁴² GEOSP is implemented on the Stanford Sherlock High-Performance Computing Cluster, which enables hybrid parallel computing with multi-threaded and distributed applications⁷⁸ on both CPU and GPU architectures. A three-dimensional benchmark case of a drained triaxial test with 1.5 million particles has been simulated using an increasing number of AMD EPYC 7543 Milan CPU cores (2, 4, 8, 16, 32, 64, and 128), each featuring 4 GB of RAM. Results are shown in Figure 8 in terms of wall clock time, which corresponds to the time required by the CPU cores

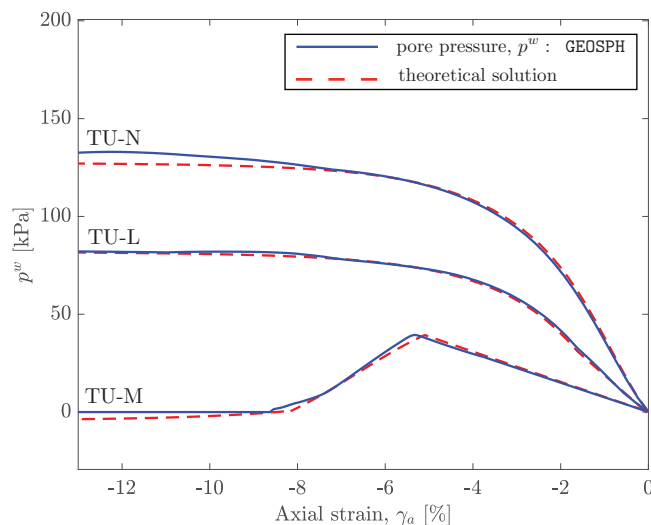


FIGURE 7 Pore-water pressure (p^w) histories for the three undrained simulations TU-M ($p'_0 = -30$ and $(p_c)_0 = -200$ kPa), TU-L ($p'_0 = -150$ and $(p_c)_0 = -200$ kPa), and TU-N ($p'_0 = -200$ and $(p_c)_0 = -200$ kPa), compared to theoretical solutions.⁷⁵

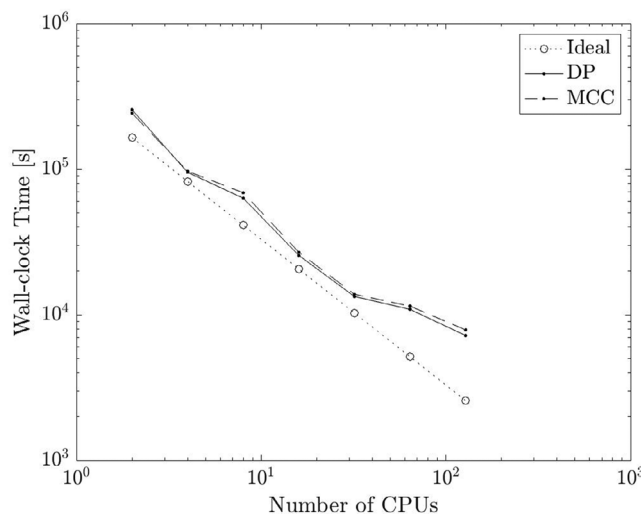


FIGURE 8 Measured wall-clock time for GEOSPH using either the Drucker-Prager (DP) or modified Cam Clay (MCC) constitutive models while simulating 1.5 million particles.

to complete the same computational job.⁷⁹ In the logarithmic scale, the measured parallel performance of the code is satisfactory compared with the theoretical performance supposing 100% efficiency.

It is worth pointing out that the proposed MCC return mapping, which bypasses the required nested Newton loops has a performance that is almost identical to that of the DP yield criterion, which has a closed form solution and does not require iterative Newton loops. Hence, Figure 8 showcases the strength of the proposed MCC formulation.

4 | REVERSE FAULT RUPTURE AND PROPAGATION UNDER DRAINED AND UNDRAINED CONDITIONS

To demonstrate the applicability of the drained-undrained SPH framework to field scale problems, the propagation of a reverse fault through a clay soil deposit modeled with the MCC criterion is simulated under drained and undrained conditions (see Table 4 for the values of MCC material parameters). The soil deposit consists of a 300-m-wide (L) and

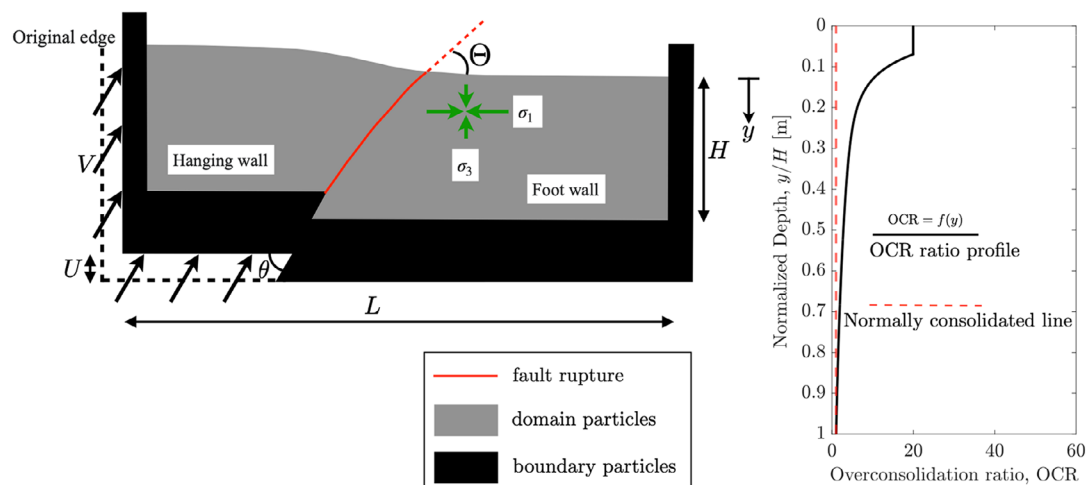


FIGURE 9 (left) Setup schematic of the fault propagation simulation, (right) the overconsolidation ratio (OCR) profile with normalized depth showing the cutoff at OCR = 20.

TABLE 3 Reverse fault propagation simulation parameters.

Parameter	Value
Time step, Δt [s]	4e-4
Initial inter-particle distance, Δ [m]	1.0
Smoothing length factor, k_h	1.5
Artificial viscosity parameters, α_π and β_π	0.2 and 0.4
Artificial stress parameter, ϵ	0.2
Damping coefficient, ξ	0.015
Pushing rate, V [m/s]	0.5
Height, H [m]	80
Length, L [m]	300
Bedrock fault orientation angle, θ [°]	60

80-m-high (H) computational domain discretized into 24,000 domain particles delimited by walls discretized into boundary particles (schematic provided in Figure 9). A reverse fault oriented at $\theta = 60^\circ$ is embedded in the rigid bedrock underlying the soil deposit which is represented by boundary particles. Fault slip is simulated by imparting a fixed velocity V on all the boundary particles surrounding the hanging wall side of the fault at the same inclination to that of the fault in the bedrock. The pushing rate was selected to be $V = 0.5$ m/s, such that the simulations avoid parasitic dynamic effects intrinsic to the SPH method, as the equivalent strain rates experienced throughout the computational domain were considerably lower than the 0.1 s^{-1} threshold at which SPH is known to exhibit certain rate-dependent behavior.⁸⁰ At the same time, the chosen pushing rate was well within the typical range of average fault slip velocities of 0.5–1.0 m/s.⁵¹ The pushing (or fault slip) is applied, until the vertical displacement of the hanging wall side U (or fault throw) attained 40% of the total height of the model ($U/H = 40\%$). Additional simulation parameters are reported in Table 3.

Prior to the initiation of the fault propagation simulation, the initial stress state is determined throughout the soil deposit. The initial stress state is crucial in geotechnical and geosciences applications, especially so in the context of the MCC model where the soil must be normally consolidated or overconsolidated, and the elastic moduli depend on the mean effective stress p' . In our simulations, the stresses are first initialized using the K_0 method, using the relation $K_0 = 1 - \sin \phi_{cs}$, and then a gravity loading is applied in conjunction to a damping term such that the particles comprising the soil deposit reached a state of geostatic stress, that is, the particles were at equilibrium in terms of a minimum kinetic energy. The initialized geostatic stress state was taken as the basis upon which to perform the faulting simulations. In the simulations employing the MCC model, a value for the initial preconsolidation pressure $(p_c)_0$ of each particle must be

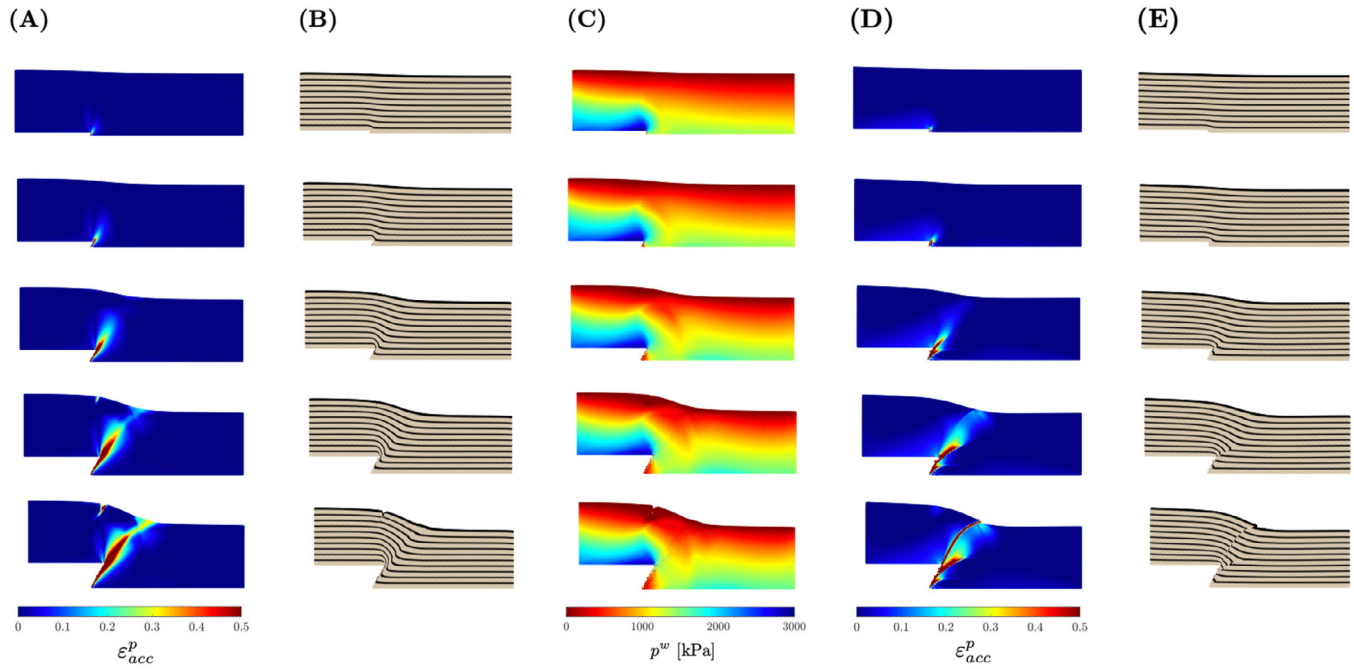


FIGURE 10 Contours of (A) the accumulated plastic strain ε_{acc}^p , (B) marker layers, and (C) pore pressure p^w for the undrained simulation, as well as (D) ε_{acc}^p , and (E) marker layers for the equivalent drained simulation at various levels of normalized vertical bedrock displacement, $U/H = 5\%, 10\%, 20\%, 30\%, 40\%$ corresponding to each row.

selected. The preconsolidation pressures are determined by assuming a distribution with depth of the overconsolidation ratio, defined as,

$$\text{OCR} = \frac{(p_c)_0}{p'_0} = f(y) \quad (42)$$

where p'_0 is the initial mean effective stress of the soil, determined at the end of the stress initialization procedure. In the simulations, a distribution of OCR ratio decreasing with depth ($f(y)$) and converging to that of a normally consolidated soil, $\text{OCR} = 1$, is used, but with a cutoff at $\text{OCR} = 20$ as the top surface is approached to avoid producing a physical strong soil at the surface (see Figure 9 for the exact depth distribution). In the case of the fault propagation simulation performed under undrained conditions, a similar stress initialization procedure is followed to the drained soil model, and the pore pressure is initialized to follow the hydrostatic gradient prior to the imposition of fault displacement and the generation of excess pore pressures.

In Figure 10, the temporal evolution of the soil deposit (depicted as a function of normalized vertical bedrock displacement U/H) as the reverse fault propagates, is displayed for both undrained and drained simulations. Contours of (A) the accumulated plastic strain ε_{acc}^p , (B) colored marker layers, and (C) pore-water pressure p^w are shown for the undrained simulations, whereas (D) the accumulated plastic strain, and (E) marker layers are shown for the drained simulations for reference. The reverse faulting simulations are characterized by the formation of a smooth rolling monocline fold and the discrete fault in the bedrock transitions into zones of localized strain or shear bands across the soil deposit. These shear bands possess certain features consistent with fault ruptures in soil, such as convex drag folding and even material layer discontinuities,³⁵ allowing them to be used as proxies for fault propagation. The amount of vertical bedrock displacement U (or the vertical component of the fault slip) required for a particular fault to reach the ground surface, known as the critical bedrock displacement U_{cr} , is an important engineering quantity as it is indicative of the relative degree of danger of a particular fault to any structures overlying or in the vicinity of the fault. Similarly, the extent of the surface region expected to be distorted by the fault known as the surface distortion zone W , is another important parameter quantifying the destructive potential of a rupture through soil cover.

In the drained simulation, localized plastic strain extends from the fault tip in the bedrock, growing upwards with increased vertical bedrock displacement. Not all the fault ruptures originating from the bedrock tip propagate the entire

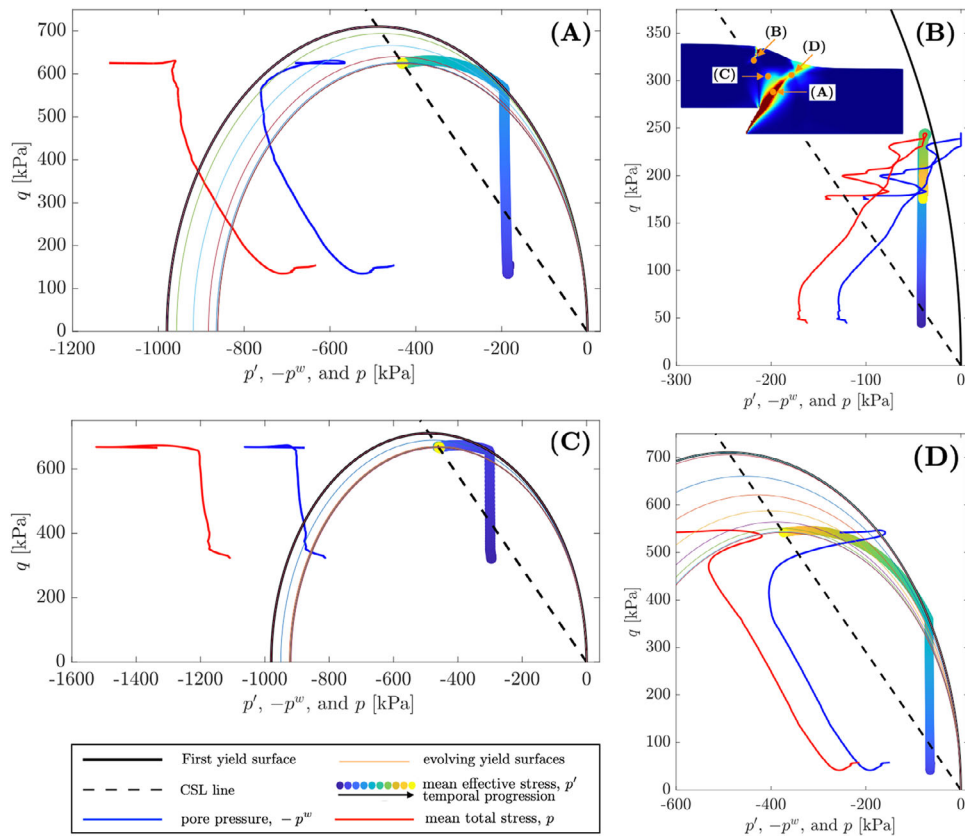


FIGURE 11 Stress paths from four measurement particles within the undrained fault propagation simulation. The locations of the particles at $U/H = 40\%$ are shown in the figure inset. Point (A) lies within the main through-going shear band at mid-height, whereas point (B) shows a particle at the tip of the tensile crack forming at the monocline hinge. Point (C) is located on the hanging wall side of the fault, and point (D) lies along the main through-going shear band near the top surface. Each panel plots the mean effective stress p' , the pore pressure ($-p^w$), and the mean total stress p versus the von Mises stress q .

way through to the top ground surface, and two ruptures remain effectively stunted. As each rupture propagates upward, its inclination shallows, and the amount of vertical fault displacement from the bedrock that can be accommodated per the amount of horizontal soil stress that must be overcome to continue propagating at a shallow angle, renders the rupture energetically unfavorable. Thus, a new steeper rupture is produced on the hanging wall side of the existing rupture. It is the second of these steeper ruptures and the third rupture in total that finally reaches the surface at $U_{cr}/H = 28.1\%$. While the majority of rupture growth occurs from the tip of the fault in bedrock, a secondary zone of strain localization emerges at the top surface and coalesces with the shear band growing from below.

The addition of excess pore pressures as a result of undrained conditions and the penalty-method approach substantially alters the behavior of the propagating fault. In the undrained case, no stunted faults occur, and a single rupture reaches the surface. In addition, the region of localized strain near the top surface propagating downwards is augmented relative to the drained simulation, although there is no visible fault scarp like in the drained simulation. Because pore pressures are highest at the bottom of the hanging wall near the tip of the fault in the bedrock, the mechanics of the hanging wall zone is most affected. Specifically, the soil here behaves in a brittle fashion, this can be seen by the formation of a tensile crack close to the monocline hinge. In centrifuge experiments of fluid-saturated clay soils, tensile cracks are also observed⁸¹ as well as in-situ in the field.⁸² The increased brittle behavior of the undrained simulation decreases the normalized critical vertical bedrock displacement required for surface rupture to occur at $U_{cr}/H = 27.0\%$, and increases the width of the surface distortion zone from $W/H = 1.03$ in the drained simulation to $W/H = 1.12$. In general, a reduced U_{cr}/H is expected for a more brittle soil as the shear wave velocity of the soil controls the rupture speed of a fault.⁴⁵ Likewise, in centrifuge experiments, U_{cr}/H was found to decrease with increased water content, and fluid-saturated soil deposits demonstrated larger W/H than dry deposits,⁸¹ corroborating the results obtained in our simulations.

In Figure 11, stress paths in $p - q$ space are shown over the duration of the undrained reverse faulting simulation at four different measurement points. All four points correctly demonstrate undrained effective stress path behavior. While the

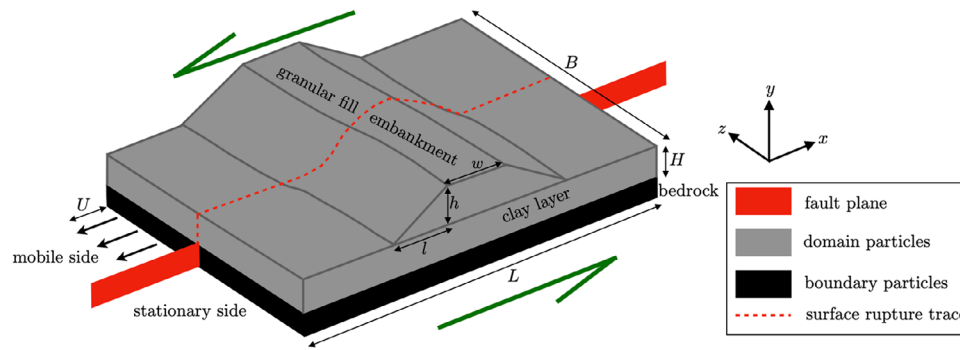


FIGURE 12 Schematic of the simulations modeling strike-slip fault rupture across an earthen embankment. Green arrows depict the direction of shear in the left-lateral strike-slip fault.

total stress paths in the undrained simulations indicate the soil deposit is being compressed, the effective stress paths lead the material to generally yield under shear as opposed to compression throughout the model. This also markedly differs from the drained tests where the material yields in a combination of shear and compression on the compression side of the CSL. Also noteworthy is that areas near the top surface of the soil deposit such as in point (B) at the tip of the tensile crack, do not plastify, and instead unload elastically as the tensile crack opens up. As can be seen, the mean effective stress remains compressive $p' < 0$, but $\sigma_{xx} > 0$ is tensile at the hinge of the monocline due to the bending deformation of the fold, leading to formation of a crack.

5 | FAULT RUPTURE THROUGH EMBANKMENTS

As a final case study, we performed two different simulations of a left-lateral strike-slip fault rupturing perpendicularly to the axis of an earthen embankment under drained and undrained loading conditions. Faults are sometimes found along the foundations of highway embankments or earth-filled dams during site investigation or even once construction has begun. This is in part due to the natural tendency of rivers and streams to establish their course following faults and due to the necessity of constructing infrastructure in seismically active regions with active fault zones.⁵³ In locations where the construction of a dam is strongly needed and the existence of active faults is known prior to any construction, embankment dams are preferable over brittle concrete dams because of the former's ability to absorb some amount of fault displacement due to the ductile nature of the constituent materials. However, there are limits to the amount of deformation that the embankment can sustain before it is breached, and the partial failure of a handful of embankments and a larger number of near-misses have occurred, for example the San Andreas and the Lower Howell dams, in the 1906 San Francisco Earthquake, the All-American Canal in the El Centro Earthquake of 1940, and the Hebgen Dam, Montana, during the West Yellowstone earthquake of 1959, to name a few.⁴⁶ In the majority of these cases the prevailing sense of slip was strike-slip as opposed to dip-slip, due to extension of strike-slip faults and their predisposition to crossing the axis of an embankment. For this reason, we focus on the rupture of strike-slip faults where the rupture occurs orthogonal to the axis of the embankment ignoring the secondary oblique slip which may have occurred in some cases in the historical record. The factors and conditions that control the amount of damage sustained by embankments under fault ruptures should be determined and analyzed. Here, we focus on the significant impact that fluid-saturated soil can have on the fault rupture process and on the generated deformation patterns throughout the embankment and the underlying soil deposit as well as on any resulting failure mechanism.

In our simulations the problem geometry is modeled as a 20 m deep homogeneous clay deposit overlying rigid bedrock (modeled as boundary particles). The clay deposit is in turn overlain by an earthen embankment made of a granular fill-like material which is 24 m high (h) and has an abutment of length 45 m (l) with a crest of width 40 m (w). The lateral dimensions of the geometry are such that the width perpendicular to the fault strike ($B = 200$ m) is greater than four times the depth of the clay layer ($H = 20$ m), as recommended by past modeling studies.⁸³ The length L of the model is 300 m. Using an initial inter-particle distance Δ of 1 m, 1,400,030 particles are used to discretize the computational domain. For a schematic figure showing the problem geometry and setup, see Figure 12. The material parameters of the granular fill and clay are shown in Table 4. In the undrained simulation, the water table is located at the top of the clay layer, and the

TABLE 4 Constitutive parameters for the different soil types in the reverse and strike-slip fault propagation simulations.

Parameter	GF	CL
Initial density, ρ [kg/m ³]	1700	1900
Peak internal friction angle, $\phi_{\max}$ [°]	34	–
Residual internal friction angle, ϕ_{res} [°]	32	–
Peak dilation angle, $\psi_{\max}$ [°]	13.0	–
Residual dilation angle, ψ_{res} [°]	0	–
Threshold octahedral plastic strain, γ_f^p	0.03	–
Cohesion, c [kPa]	30	–
Young's modulus, E [MPa]	20	–
Poisson ratio, ν	0.3	0.3
Initial OCR ratio,	$f(y)$	$f(y)$
Critical state line slope, M	–	1.45
Initial void ratio, e	–	0.7
Virgin isotropic consolidation index, λ	–	0.35
Swelling index, κ	–	0.05

Note: CL stands for clay modeled with MCC whereas GF refers to the granular fill material modeled with DP.

clay deposit is assumed to be fully saturated, while the overlying embankment is dry. In the simulation, a strike-slip fault located within the bedrock propagates for a total normalized slip of $U/H = 45\%$ at a rate of 0.5 m/s, consistent with the range of slips and slip velocity observed in the field for strike-slip type faults. The slippage of the fault in bedrock, in turn, drags along and deforms the overlying soil.

To simulate the fault rupture, a velocity orthogonal to the axis of the embankment is applied to half of the model such that a left-lateral strike slip fault traverses the soil deposit and the embankment perpendicularly to its axis. Specifically, the velocity is applied to the boundary particles along the bottom boundary of the soil deposit corresponding to the side of the model that is displaced, to allow the soil to deform without any constraints above the bedrock. Periodic boundary conditions are applied in the x -direction parallel to the strike of the fault (perpendicular to the fault axis), to avoid any boundary effects caused by imposing rigid boundary conditions in the y -direction as is common in rigid shear box setups for analogue or centrifuge experiments.

Visualizations of undrained and drained simulations are displayed in Figure 13, showing three-dimensional contour plots of the accumulated plastic strain (ϵ_{acc}^p) in panels (A) and (B) respectively at the end of the simulations after a normalized fault slippage of $U/H = 45\%$. Panels (C) and (D) present a bird's-eye view (projection on the x - z plane) looking downwards along the y -direction, whereas panels (E) and (F) show the pore pressure p^w and accumulated plastic strain for the undrained simulation on the x - z plane but from the bottom of the model looking upwards in the y -direction.

In the drained model as slip begins to accumulate along the strike-slip fault, strain is localized into a narrow band in the soil just above the fault in the bedrock, and rapidly extends upwards towards the soil deposit top surface. Throughout the clay soil deposit, the band mostly propagates vertically upward with some slight oscillating lateral deflection, indicating that the through-going shear band is likely the result of an amalgamation of smaller en-echelon Riedel type shears.^{83,84} As the shear band propagates into the overlying embankment, its trajectory is significantly altered by the surface profile of the embankment and the principal stress direction aligned with the abutment slopes leading to a characteristically curved S-shaped surface rupture trace (panel D in Figure 13), also observed in 1-g analogue experiments.⁸⁵

The undrained simulation initiates similarly, with a shear band cutting across the model directly above the strike-slip fault in the bedrock and rapidly propagating across the clay deposit, mainly forward of the embankment ($x = -50$ to -150 m). However, different from the drained model, excess pore pressures rapidly build up along the base of the soil deposit facing the bedrock, particularly below the embankment and on the side where the bedrock is mobilized. The pore pressure radiates from the fault plane over continued slip (see bottom row of Figure 14 for views of the base). After around $U/H = 10\%$, strain localizes perpendicular to fault strike along the base of the soil deposit roughly below the toe of the abutment of the embankment forming a protrusion from the main shear band above the fault (see second to bottom row of Figure 14). By $U/H = 15\%$ of normalized slip, a failure surface has developed as additional localized strain propagates from this accumulated plastic strain protrusion outcropping along the embankment abutment surface on the opposing side of the embankment following a log-spiral trajectory (see first row of Figure 14 for an illustrative

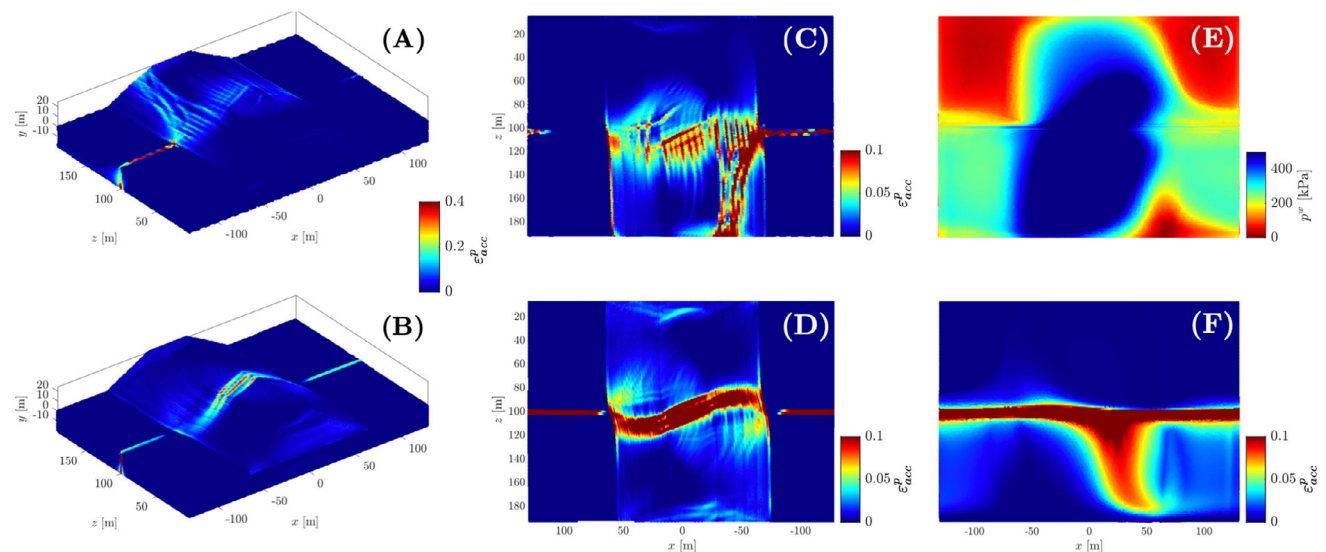


FIGURE 13 Accumulated plastic strain (ε_{acc}^p) contours of the embankment and soil deposit after $U/H = 45\%$ of strike-slip fault rupture perpendicular to the embankment axis for the (A) undrained, and (B) drained simulations. Panels (C) and (D) show bird's-eye views of (A) and (B) respectively, and (E) and (F) show the pore pressure (p^w) and ε_{acc}^p , respectively, after $U/H = 45\%$ for the undrained simulation looking at the bottom of the model with the vertical y -direction into the page.

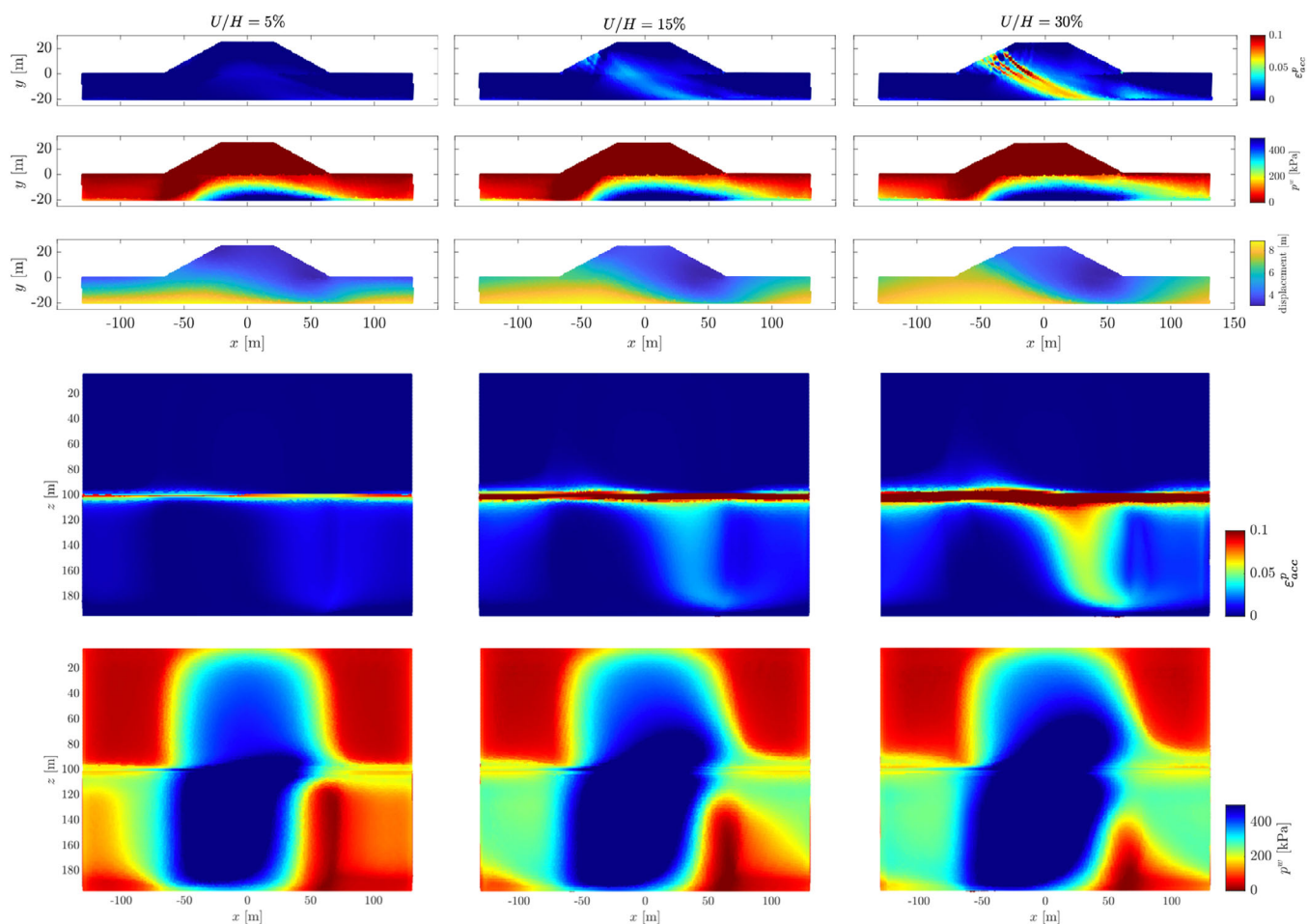


FIGURE 14 Temporal evolution of ε_{acc}^p , p^w , and displacement for the strike-slip fault rupture simulation under undrained conditions at three different amounts of normalized fault slip (U/H). Top three rows show a cross-section at $z = 160$ m, whereas the bottom two rows display the model looking from the bottom upwards along the vertical y -direction into the page and plot ε_{acc}^p and p^w .

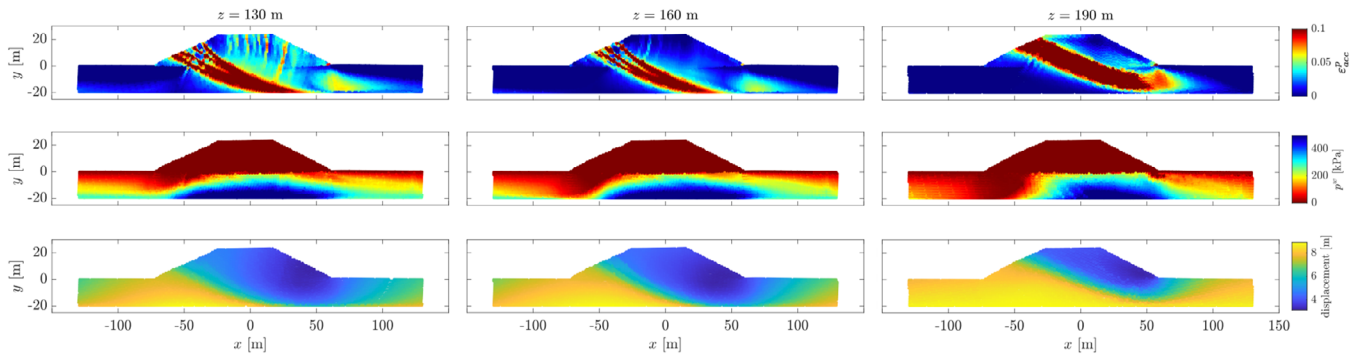


FIGURE 15 Cross sections of the embankment and soil deposit parallel to fault strike, showing ϵ_{acc}^p (top row), p^w (middle row), and displacement (bottom row), at the final normalized fault slip of $U/H = 45\%$.

cross section parallel to fault strike at $z = 160$ m). The failure surface originates from the initial localized plastic strain at the protrusion perpendicular to the fault strike below the abutment edge on the soil deposit bottom, as the pore pressure is still quite high here (see Figure 14 second row), but the mean effective stress is low compared to directly below the embankment. Hence, conditions for initial failure are optimal here. As slip further accumulates on the fault, the localized accumulated plastic strain composing the slope failure surface increases in magnitude and while the abutment itself does not fail completely by $U/H = 45\%$, a portion of the embankment delimited by the slope failure surface displaces in the direction opposite the fault displacement, as a result of sliding along the failure surface slip plane, reducing the overall displacement of the affected sections of the embankment and the adjoining soil. This process leads to an overall reduction in accumulated plastic strain on the backward side of the embankment surface, and does not allow for the distinctive S-shaped surface trace of the fault to develop as seen in Figure 13, panel (C), and the magnitude of ϵ_{acc}^p is also reduced along the main shear through-going band along its surface rupture trace at the top surface of the soil deposit.

Comparing the top surfaces of the simulation model in panels (C) and (D) of Figure 13, it can be seen that the embankment suffers less shear deformation in the undrained than drained simulations by noting the greater lateral displacement in the edges of the embankment abutments in the drained simulation. Here again, the sliding from the developing failure surface acts opposite the displacement from the fault. In addition, looking at panel (C), the surface projection of the failure surface emanating from the base and a series of small secondary vertically oriented shear bands manifest themselves as shear bands roughly parallel to the embankment axis, mostly on the mobilized side of the model. Cross sections along the z -direction parallel to fault strike in Figure 15, help to visualize the extent and depth of the slope failure surface that emerges in the undrained simulation. While the embankment is exposed to less shear offset resulting directly from the strike-slip fault in the undrained simulation compare to the drained case, the amount of plastic deformation in the interior of the embankment due to the slope failure surface puts its integrity at risk as well.

A commonly used metric to evaluate the functionality of an embankment after an event such as the rupture of a nearby or traversing fault is the relative compaction (RC) which is the ratio between the in-situ dry density of the granular fill to the maximum dry density of the same material. The relative compaction between two states, initial RC_i and final RC_f , can be related to the change in volumetric strain of the soil $\Delta\epsilon_v$ between those two states as follows,

$$\Delta\epsilon_v = \frac{RC_i}{RC_f} - 1. \quad (43)$$

In general, relative compaction of $RC < 94\%$ can be considered a threshold where the fill material is unsuitable and the embankment must be repaired.⁸⁶ Assuming an initial $RC = 95\%$ after stress initialization prior to fault loading, which is the typical properly compacted in-situ value for a granular fill embankment, the final relative compaction RC_f at the end of the simulation or at any point in time can be determined. In Figure 16, the regions of the embankment where $RC < 94\%$, requiring repair are extracted from the simulation model for inspection in the (A) undrained, and (B) drained simulations at maximum slip, whereas the evolution in time (as a function of normalized fault slip) of the total percentage of the embankment that is damaged and necessitating repair is plotted in Figure 17 for both simulations.

By $U/H = 45\%$, the embankment subjected to fault rupture under undrained conditions possesses a larger percentage that is damaged, at 22.36%, compared to 18.24% of the drained simulation. In the undrained simulation the damaged zone follows the shear bands pertaining to the slope failure surface on the side of the model that is mobilized, whereas in the

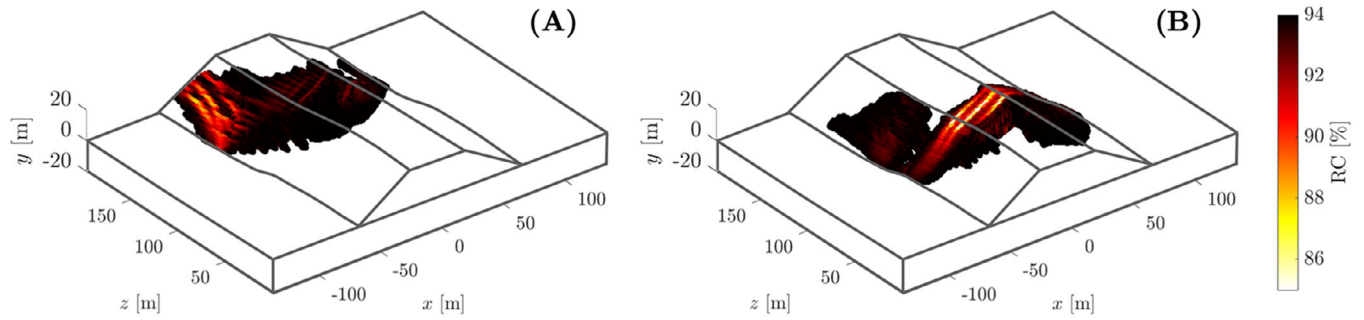


FIGURE 16 Extraction of the damaged zone of the embankments ($RC < 94\%$, shown as contours superimposed over the outline of the computational model) after fault rupture at a maximum normalized slip of $U/H = 45\%$ in the (A) undrained, and (B) drained simulations.

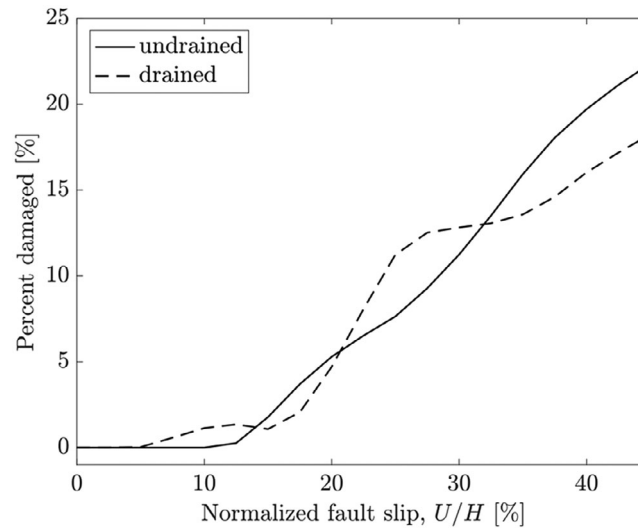


FIGURE 17 Temporal evolution of the percentage of the embankment that is damaged ($RC < 94\%$) as a function of normalized fault slip in the undrained and drained models.

drained case the damaged zone consists of the shear band that ruptures through the middle of the model projecting in an S-shape at the top surface. In both cases, particles within shear bands and zones of localized strain which also happen to be at or near the top surface exhibit the lowest relative compaction, as a result of the dilation that occurs in the shear band and of the low confining pressure. Although the final percent damage is greater in the undrained simulation, it is the drained simulation which sustains relative compaction below the 94% threshold first after only $U/H = 5\%$ of normalized fault slip, and also experiences greater percent damage between $U/H = 20\%$ to $U/H = 30\%$.

6 | DISCUSSION AND CONCLUSION

We presented a new framework for simulating undrained loading of saturated soils employing smoothed particle hydrodynamics (SPH), based on a one-layer two-phase formulation, and the modified Cam Clay (MCC) constitutive model. The framework is shown to perform very well against benchmark problems some of which, most notably the triaxial tests, are relatively non-standard practice in the literature due to the difficulty of applying confining boundary conditions in SPH. Then, significantly, our framework is applied to investigate two scenarios involving undrained loading conditions and large deformations which are suited to the abilities of the proposed framework. The framework demonstrates its efficacy in examining reverse faulting through a clay deposit (Section 4) and the rupture of a strike-slip fault across a clay deposit overlain by a granular fill embankment (Section 5). Through these explorations, an array of valuable practical insights and novel perspectives emerge, which reshape considerations related to embankment stability and safety. Notably, previous

computational modeling efforts have predominantly addressed these issues under dry or drained conditions, thus our work emphasizes the importance of incorporating pore fluid pressure into simulations of saturated soil.

In the simulations of reverse fault rupture through clay soil cover, our results indicate that excess pore pressure in the saturated soil leads the soil to behave in a brittle manner, reducing the critical amount of slip needed for surface rupture, and creating small tension cracks along the monocline hinge line, which agrees with analogue experiments using wet clays⁸¹ and field observations.⁸² Furthermore, the width of the surface distortion zone, that is, the extent along the soil top surface deformed by the propagating fault, increases relative to the dry simulations. These findings suggest that soil saturation by pore fluid can increase the risk posed by propagating faults to any overlying structure, as damage to the surface will appear with more commonly occurring seismic events of smaller magnitude, and the affected surface area will also increase.

In the simulations of strike-slip rupture perpendicular to the axis of an embankment, the saturation of the bottom clay layer leads to excess pore pressures which trigger a slope failure within the abutment of the embankment causing the embankment to slide in the direction opposite that of fault displacement. This process reduces the deformation directly caused by the strike-slip fault while increasing the overall deformation and damage sustained by the embankment. For example, the amount of shear offset across the fault is reduced, and the lack of a large and clear surface rupture trace or scarp at the top of the embankment from the strike-slip fault could be misleading, underrepresenting the true extent of damage in the embankment, as the majority of the damage in the undrained simulation lies in its interior within the large shear band that accompanies the slope failure surface that develops. The extensive dilation suffered by the soil in these shear zones which are less visible from the exterior of the embankment would likely require re-compaction or complete removal and replacement.

The proposed framework does not include modeling of transient effects relating to pore fluid such as fluid flow through the porous networks of the soil, or the dissipation of the excess pore pressures,^{87,88} as well as the effect of partial saturation.^{89,90} These processes could impact the long-term performance and potential settlement of the damaged embankment, most notably if it is not immediately repaired. The undrained framework for saturated soils also relies on a constitutive model that can produce volumetric plastic strains. Otherwise there will be no excess pore pressures generated after the material plastifies. This fact obliges the use of constitutive models similar to MCC whenever the soil is saturated. Our penalty-method based formulation is also not intended to model pore pressures associated with suction, which may be relevant in problems involving extensive dilation, or when modeling the pullout capacity of suction anchors under undrained conditions. Future work seeking to model suction or transient effects would likely require employing a fully coupled formulation based on the Biot-Zienkiewicz equations using mixture theory such as the $\mathbf{u} - p$ or $\mathbf{u} - \mathbf{w} - p$ formulations. The latter formulation would be particularly pertinent in applications where the relative acceleration between the pore water and soil phases cannot be ignored. In the context of embankment stability in the wake of fault rupture, an analysis involving transient effects could be valuable given the considerable amount of time it may take for civil authorities to repair damaged infrastructure of secondary importance such as a road embankment when they may deceptively seem mostly still viable from the surface, as demonstrated in the undrained simulations for saturated soil performed in this paper. Nevertheless, with a computationally less costly framework, we have shown how valuable knowledge can be gained from the extreme limit cases of drained and undrained loading conditions which can be useful in a wide range of problems in geotechnics, and which in the case of fault rupture are especially pertinent in understanding the propagation process.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

1. Borja RI. Free boundary, fluid flow, and seepage forces in excavations. *J Geotech Eng*. 1992;118(1):125-146.
2. Borja RI. Analysis of incremental excavation based on critical state theory. *J Geotech Eng*. 1990;116(6):964-985.
3. Abe K, Soga K, Bandara S. Material point method for coupled hydromechanical problems. *J Geotech Geoenviron Eng*. 2014;140(3):04013033.
4. Kularathna S, Liang W, Zhao T, Chandra B, Zhao J, Soga K. A semi-implicit material point method based on frictional-step method for saturated soil. *Int J Numer Anal Methods Geomech*. 2021;45(2):1405-1436. doi:10.1002/nag.3207
5. Wu F, Sin W, Li X, Guan Y, Dong M. Material point method-based simulation of dynamic process of soil landslides considering pore fluid pressure. *Int J Numer Anal Methods Geomech*. 2023;47(13):2385-2404.
6. Sun W, Fish J. Coupling of non-ordinary state-based peridynamics and finite element method for fracture propagation in saturated porous media. *Int J Numer Anal Methods Geomech*. 2021;45:1260-1281.
7. Mohajerani S, Wang G, Zhao Y, Jin F. A novel peridynamics modelling of cemented granular materials. *Acta Geotech*. 2023;18:2529-2548.
8. Zhou X, Zhang T. Generalized plastic ordinary state-based peridynamic model with shear deformation of geomaterials. *Acta Geotech*. 2023;18:609-623.
9. Oñate E, Idelsohn SR, Pin FD, Aubry R. The particle finite element method. An overview. *Int J Comput Methods*. 2004;1(2):267-307.
10. Monforte L, Arroyo M, Carbonell JM, Gens A. Numerical simulation of undrained insertion problems in geotechnical engineering with the Particle finite element method (PFEM). *Comput Geotech*. 2017;82:144-156.
11. Wang L, Zhang X, Tinti S. Large deformation dynamic analysis of progressive failure in layered clayey slopes under seismic loading using the particle finite element method. *Acta Geotech*. 2021;16:2435-2448. doi:10.1007/s11440-021-01142-8
12. Belytschko T, Lu YY, Gu L. Element-free Galerkin methods. *Int J Numer Methods Eng*. 1994;37(2):229-256.
13. Wu CT, Chen JS, Chi L, Huck F. Lagrangian meshfree formulation for analysis of geotechnical materials. *J Eng Mech*. 2001;127(5):440-449.
14. Samimi S, Pak A. Three-dimensional simulation of fully coupled hydro-mechanical behavior of saturated porous media using element free Galerkin (EFG) method. *Comput Geotech*. 2012;46:75-83.
15. Liu WK, Jun S, Zhang YF. Reproducing kernel particle methods. *Int J Numer Methods Fluids*. 1995;20(8-9):1081-1106.
16. Siriakorn T, Chi SW, Foster C, Mahdavi A. u-p semi-Lagrangian reproducing kernel formulation for landslide modeling. *Int J Numer Anal Methods Geomech*. 2018;42(2):209-376.
17. Wei H, Chen JS, Beckwith F, Baek J. A naturally stabilized semi-Lagrangian meshfree formulation for multiphase porous media with application to landslide modeling. *J Eng Mech*. 2020;146(4):04020012.
18. Bui HH, Fukagawa R, Sako K, Wells JC. Slope stability analysis and discontinuous slope failure simulation by elasto-plastic smoothed particle hydrodynamics (SPH). *Géotechnique*. 2011;61(7):565-574.
19. Li Y, Liu HQ, Yang L, Liu Y. An optimized DEM-SPH model for surge waves induced by riverside landslides. *Int J Numer Anal Methods Geomech*. 2023;48(1):270-286.
20. Wang Y, Wu W, Peng C. Regularized SPH model for soil-structure interaction with generalized frictional boundary. *Int J Numer Anal Methods Geomech*. 2023;47(10):1852-1875.
21. Lucy LB. A numerical approach to the testing of the fission hypothesis. *Astron J*. 1977;82(12):1013.
22. Gingold R, Monaghan J. Smoothed particle hydrodynamics: theory and application to nonspherical stars. *Mon Not R Astron Soc*. 1977;181:375-389.
23. Nonoyama H, Moriguchi S, Sawada K, Yashima A. Slope stability analysis using smoothed particle hydrodynamics (SPH) method. *Soils Found*. 2015;55(2):458-470.
24. Zhou Q, Xu WJ, Dong XY. SPH-DEM coupling method based on GPU and its application to the landslide tsunami. Part I: method and validation. *Acta Geotech*. 2022;17:2101-2119.
25. Xu WJ, Zhou Q, Dong XY. SPH-DEM coupling method based on GPU and its application to the landslide tsunami. Part II: reproduction of the Vajont landslide tsunami. *Acta Geotech*. 2022;17:2121-2137.
26. Gao L, Pastor M, Li T, et al. A framework coupled neural networks and SPH depth integrated model for landslide propagation warning. *Acta Geotech*. 2023;18:3863-3888.
27. Pastor M, Haddad B, Sorbino G, Cuomo S, Drempetic V. A depth-integrated, coupled SPH model for flow-like landslides and related phenomena. *Int J Numer Anal Methods Geomech*. 2009;33:143-172.
28. Pastor M, Yague A, Stickle MM, Manzanal D, Mira P. A two-phase SPH model for debris flow propagation. *Int J Numer Anal Methods Geomech*. 2017;42:418-448.
29. Fávero Neto AH, Askarinejad A, Springman SM, Borja RI. Simulation of debris flow on an instrumented test slope using an updated Lagrangian particle method. *Acta Geotech*. 2020;15:2757-2777.
30. Pastor M, Tayyebi SM, Stickle MM, et al. An arbitrary Lagrangian eulerian (ALE) finite difference (FD)-SPH depth integrated model for pore pressure evolution on landslides over erodible terrains. *Int J Numer Anal Methods Geomech*. 2022;46(6):1127-1153.
31. Fávero Neto AH, Borja RI. Continuum hydrodynamics of dry granular flows employing multiplicative elastoplasticity. *Acta Geotech*. 2018;13:1027-1040.
32. Nguyen NHT, Bui HH, Nguyen GD. Effects of material properties on the mobility of granular flow. *Granular Matter*. 2020;22(59).
33. Yang E, Bui H, Sterck HD, Nguyen G. A scalable parallel computing SPH framework for predictions of geophysical granular flows. *Comput Geotech*. 2020;121:103474.
34. Xiong H., Hao M., Zhao D., Qiu Y., Chen X. Study of the dynamics of water-enriched debris flow and its impact on slit-type barriers by a modified SPH-DEM coupling approach. *Acta Geotech*. 2023;19(2):1-27. doi:10.1007/s11440-023-02106-w

35. del Castillo EM, Fávero Neto AH, Borja RI. Fault propagation and surface rupture in geologic materials with a meshfree continuum method. *Acta Geotech*. 2021;16:2463-2486. doi:10.1007/s11440-021-01233-6
36. del Castillo EM, Fávero Neto AH, Borja RI. A continuum meshfree method for sandbox-style numerical modeling: application to accretionary and doubly vergent wedges. *J Struct Geol*. 2021;153:104466.
37. Mu D, Tang A, Li Z, Qu H, Huang D. A bond-based smoothed particle hydrodynamics considering frictional contact effect for simulating rock fracture. *Acta Geotech*. 2023;18:625-649.
38. del Castillo EM, Fávero Neto AH, Borja RI. Modeling fault rupture through layered geomaterials with SPH. In: Dyskin A, Pasternak E, eds. *Multiscale Processes of Instability, Deformation and Fracturing in Geomaterials*. Springer Series in Geomechanics and Geoengineering. IWBBDG 2022. Springer; 2023. doi:10.1007/978-3-031-22213-9_16
39. Morikawa DS, Asai M. Soil-water strong coupled ISPH based on u-w-p formulation for large deformation problems. *Comput Geotech*. 2022;142(104570):104570.
40. Lian Y, Bui HH, Nguyen GD, Zhao S, Haque A. A computationally efficient SPH framework for unsaturated soils and its application to predict the entire rainfall-induced slope failure process. *Géotechnique*. 2022;1751-7656. doi:10.1680/jgeot.21.00349
41. Lian Y, Bui HH, Nguyen GD, Haque A. An effective and stabilised ($u - p_l$) SPH framework for large deformation and failure analysis of saturated porous media. *Comput Meth Appl Mech Eng*. 2023;408:115967.
42. Ramachandran P, Bhosale A, Puri K, et al. PySPH: a python-based framework for smoothed particle hydrodynamics. *ACM Trans Math Softw (TOMS)*. 2021;47(4):1-38.
43. Fávero Neto AH. *A Continuum Lagrangian Finite Deformation Computational Framework for Modeling Granular Flows*. PhD thesis. Stanford University; 2020.
44. Mullet B, Segall P, Fávero Neto AH. Numerical modeling of caldera formation using smoothed particle hydrodynamics (SPH). *Geophys J Int*. 2023;234(2):887-902.
45. Loukidis D, Bouckovalas GD, Papadimitriou AG. Analysis of fault rupture propagation through uniform soil cover. *Soil Dyn Earthquake Eng*. 2009;29(11-12):1389-1404.
46. Bray JD. *The Effects of Tectonic Movements on Stresses and Deformations in Earth Embankments*. PhD thesis. University of California at Berkeley; 1990.
47. Bray JD, Seed RB, Cluff LS, Seed HB. Earthquake fault rupture propagation through soil. *J Geotech Eng*. 1994;120(3):543-561.
48. Anastasopoulos I, Gazetas G. Foundation-structure systems over a rupturing normal fault: Part II. Analysis of the Kocaeli case histories. *Bull Earthquake Eng*. 2007;5:277-301.
49. Thebian L, Najjar S, Sadek S, Mabsout M. Numerical investigation of dip-slip fault propagation effects on offshore seabed sediments. *Eng Geol*. 2018;237:149-167.
50. Garcia FE, Bray JD. Distinct element simulations of shear rupture in dilatant granular media. *Int J Geomech*. 2018;18(9):04018111.
51. Oettle NK, Bray JD, Dreger DS. Dynamic effects of surface fault rupture interaction with structures. *Soil Dyn Earthquake Eng*. 2015;72:37-47.
52. Hazeghian M, Soroush A. DEM simulation of reverse faulting through sands with the aid of GPU computing. *Comput Geotech*. 2015;66:253-263.
53. Sherard JL, Cluff LS, Allen CR. Potentially active faults in dam foundations. *Géotechnique*. 1974;24(3):367-428.
54. Bray JD, Seed RB, Seed HB. 1 g Small-scale modelling of saturated cohesive soils. *Geotech Test J*. 1993;16(1):46-53.
55. Lazarte CA. *The Response of Earth Structures to Surface Fault Rupture*. PhD thesis. University of California at Berkeley; 1996.
56. Borja RI. Cam-clay plasticity. Part V: a mathematical framework for three-phase deformation and strain localization analyses of partially saturated porous media. *Comput Meth Appl Mech Eng*. 2004;193(48-51):5301-5338.
57. Borja RI. On the mechanical energy and effective stress in saturated and unsaturated porous continua. *Int J Solids Struct*. 2006;43(6):1764-1786.
58. Borja RI. *Plasticity Modeling & Computation*. Springer; 2013.
59. Dienes J. On the analysis of rotation and stress rate in deforming bodies. *Acta Mech*. 1979;32:217-232.
60. Borja RI, Tamagnini C. Cam-clay plasticity Part III: extension of the infinitesimal model to include finite strains. *Comput Meth Appl Mech Eng*. 1998;155:73-95.
61. Bui HH, Nguyen GD. Smoothed particle hydrodynamics (SPH) and its applications in geomechanics: from solid fracture to granular behavior and multiphase flows in porous media. *Comput Geotech*. 2021;138:104315.
62. Dyka CT, Randles PW, Ingle RP. Stress points for tension instability in SPH. *Int J Numer Methods Eng*. 1997;40:2325-2341.
63. Bui HH, Fukagawa R, Sako K, Ohno S. Lagrangian meshfree particles method (SPH) for large deformation and failure flows of geomaterial using elastic-plastic soil constitutive model. *Int J Numer Anal Methods Geomech*. 2008;32:1537-1570.
64. Liu G, Liu M. *Smoothed Particle Hydrodynamics: A Meshfree Particle Method*. World Scientific Co.; 2003.
65. Violeau D. *Fluid Mechanics and the SPH Method*. Oxford University Press; 2012.
66. Gray J, Monaghan J, Swift R. SPH elastic dynamics. *Comput Meth Appl Mech Eng*. 2001;190(49-50):6641-6662.
67. Adami S, Hu XY, Adams NA. A generalized wall boundary condition for smoothed particle hydrodynamics. *J Comput Phys*. 2012;231:7057-7075.
68. Roscoe KH, Burland JH. On the generalized stress-strain behavior of 'wet' clay. In: Heyman J, Leckie FA, eds. *Engineering Plasticity*. Cambridge University Press; 1968:535-609.
69. Borja RI, Tamagnini C, Amorosi A. Coupling plasticity and energy-conserving elasticity models for clays. *J Geotech Geoenviron Eng*. 1997;123(10):948-957.

70. Borja RI, Lee SR. Cam-clay plasticity, Part I: implicit integration of elasto-plastic constitutive relations. *Comput Meth Appl Mech Eng*. 1990;78(1):49-72.
71. Hughes T, Winget J. Finite rotation effects in numerical integration of rate constitutive equations arising in large-deformation analysis. *Int J Numer Methods Eng*. 1980;15(12):1862-1867.
72. Chen X, Leung Y, Mori H, Uchida S, Takumi K. Single-layer soil-water coupled SPH method and its application to sinkhole simulation. *Acta Geotech*. 2023;19(3):991-1018.
73. Pereira GG, Cleary PW, Lemiale V. SPH method applied to compression of solid materials for a variety of loading conditions. *Appl Math Modell*. 2017;44:72-90.
74. Zhao S, Bui HH, Lemiale V, Nguyen GD, Darve F. A generic approach to modelling flexible confined boundary conditions in SPH and its application. *Int J Numer Anal Methods Geomech*. 2019;43(5):1005-1031.
75. Wood DM. *Soil Behavior and Critical State Soil Mechanics*. Cambridge University Press; 1990.
76. Crespo AJ, Domínguez JM, Rogers BD, et al. DualSPHysics: open-source parallel CFD solver based on smoothed particle hydrodynamics (SPH). *Comput Phys Commun*. 2015;187:204-216.
77. Bui HH, Nguyen GD. Smoothed particle hydrodynamics (SPH) and its applications in geomechanics: from solid fracture to granular behaviour and multiphase flows in porous media. *Comput Geotech*. 2021;138:104315.
78. Oger G, Le Touzé D, Guibert D, et al. On distributed memory MPI-based parallelization of SPH codes in massive HPC context. *Comput Phys Commun*. 2016;200:1-14.
79. Ferrari A, Dumbser M, Toro EF, Armanini A. A new 3D parallel SPH scheme for free surface flows. *Comput Fluids*. 2009;38(6):1203-1217.
80. Wang Y, Bui HH, Nguyen GD, Ranjith PG. A new SPH-based continuum framework with an embedded fracture process zone for modelling rock fracture. *Int J Solids Struct*. 2019;159:40-57.
81. Ahmadi M, Moosavi M, Jafari MK. Experimental investigation of reverse fault rupture propagation through wet granular soil. *Eng Geol*. 2018;239:229-240.
82. Plafker G. Surface faults on Montague island associated with the 1964 Alaska earthquake. *US Geol Surv Prof Pap*. 1967;543-G:G1-G42.
83. Agalianos A, Coquereau d. C. dO, Anastasopoulos I. Rigid slab/foundation subjected to strike-slip faulting: mechanisms and insights. *Géotechnique*. 2020;70(4):354-373.
84. Taniyama H. Numerical analysis of overburden soil subjected to strike-slip fault: distinct element analysis of Nojima fault. *Eng Geol*. 2011;123(3):194-203.
85. Lazarte CA, Bray JD. A study of strike-slip faulting using small-scale models. *Geotech Test J*. 1996;19(2):118-129.
86. Petela E, Klimis NS. Numerical assessment of distortion and damage induced on a highway embankment due to reverse fault rupture propagation. *Soil Dyn Earthquake Eng*. 2019;116:264-277.
87. Borja RI, Alarcón E. A mathematical framework for finite strain elastoplastic consolidation Part 1: balance laws, variational formulation, and linearization. *Comput Meth Appl Mech Eng*. 1995;122(1-2):145-171.
88. Borja RI, Tamagnini C, Alarcón E. Elastoplastic consolidation at finite strain part 2: finite element implementation and numerical examples. *Comput Meth Appl Mech Eng*. 1998;159(1-2):103-122.
89. Song X, Borja RI. Mathematical framework for unsaturated flow in the finite deformation range. *Int J Numer Methods Eng*. 2014;97(9):658-682.
90. Song X, Borja RI. Finite deformation and fluid flow in unsaturated soils with random heterogeneity. *Vadose Zone J*. 2014;13(5):1-11. doi:10.2136/vzj2013.07.0131

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APPENDIX A: MCC RETURN MAPPING ALGORITHM

Here we provide additional details on the approximate closest point projection (CPP) return mapping algorithm for the modified Cam Clay (MCC) model. It is summarized in Algorithm A1. To determine the derivative of the yield function with respect to the consistency parameter $\partial \mathcal{F} / \partial \Delta \phi$, needed in the local Newton iteration (Equation 35), the derivatives of the von Mises stress q , mean effective stress p' , and preconsolidation pressure p_c , with respect to the consistency parameter must be determined as they are linked via the following expression,

$$\frac{\partial \mathcal{F}}{\partial \Delta \phi} = (2p' - p_c) \frac{\partial p'}{\partial \Delta \phi} + \left(\frac{2q}{M^2} \right) \frac{\partial q}{\partial \Delta \phi} - p' \frac{\partial p_c}{\partial \Delta \phi}, \quad (\text{A1})$$

and are written below:

ALGORITHM A1 MCC approximate CPP return mapping algorithm.**Input:** $K_n, \mu_n, \sigma'_n, p'^{\text{tr}}, q^{\text{tr}}, (p_c)_n, v_n,$ **Output:** $K_{n+1}, \mu_{n+1}, \sigma'_{n+1}, (p_c)_{n+1}, v_{n+1}$ Set initial guesses: $\Delta\phi^0 = 0, p'^0 = p'^{\text{tr}}, q^0 = q^{\text{tr}},$ and $(p_c)^0 = (p_c)_n$ 1. Evaluate $\partial q / \partial \Delta\phi$ (Equation A2), $\partial p' / \partial \Delta\phi$ (Equation A3), $\partial p_c / \partial \Delta\phi$ (Equation A4).2. Calculate $\partial F / \partial \Delta\phi$ (Equation A1).3. Update the consistency parameter $\Delta\phi^{k+1}$ (Equation 35).3. Update the preconsolidation pressure $(p_c)^k$, solving Equation A6.4. Update the mean effective stress p'^k (Equation 33) and deviatoric stress q^k (Equation 34). $k \leftarrow k + 1$ Set $p'_{n+1} = p'^k, q_{n+1} = q^k, (p_c)_{n+1} = (p_c)^k.$ Update $e_{n+1}, v_{n+1}, K_{n+1}$ and μ_{n+1} (Equation 29).Determine σ'_{n+1} and time integrate to find σ'_{n+1} from Equations (37) and (38).

$$\frac{\partial q}{\partial \Delta\phi} = -\frac{q}{\Delta\phi + M^2/6\mu} \quad (\text{A2})$$

$$\frac{\partial p'}{\partial \Delta\phi} = \frac{-2K\left(\Delta\phi(2p' - p_c) + \frac{1}{v}\right)^2\left(p' - \frac{p_c}{2}\right)}{a} \quad (\text{A3})$$

and

$$\frac{\partial p_c}{\partial \Delta\phi} = \frac{(-2p' + p_c)\frac{(p_c)_n}{v}}{a} \quad (\text{A4})$$

with,

$$a = 8K\left(p' - \frac{p_c}{2}\right)^2 \Delta\phi^3 + 8\left(\frac{K}{v} + \frac{p'}{2} - \frac{p_c}{4}\right)\left(p' - \frac{p_c}{2}\right)\Delta\phi^2 + \frac{2}{v}\left(\frac{K}{v} + 2p' - p_c - \frac{(p_c)_n}{2}\right)\Delta\phi + \left(\frac{1}{v}\right)^2. \quad (\text{A5})$$

An additional feature of the approximate CPP algorithm is that for the preconsolidation pressure update at the k th iteration, $(p_c)^k$, the following quadratic equation must be solved by picking the negative root for $(p_c)_{n+1}$,

$$((p_c)_{n+1})^2 - \frac{b}{\Delta\phi}(p_c)_{n+1} + \frac{c}{\Delta\phi} = 0 \quad (\text{A6})$$

with $b = (1 + 2K\Delta\phi)/v + 2\Delta\phi p'^{\text{tr}}$ and $c = (p_c)_n(1 + 2K\Delta\phi)/v$.

APPENDIX B: ADDITIONAL FRAMEWORK VERIFICATION

Here we include the visualizations and results of additional verification tests for the modified Cam Clay (MCC) model and the penalty-method based framework in SPH. Figures B1, B2, B3, B4, B5, and B6 are all referenced in the main body of the paper and display results from a series of simple shear tests and triaxial tests under drained and undrained conditions.

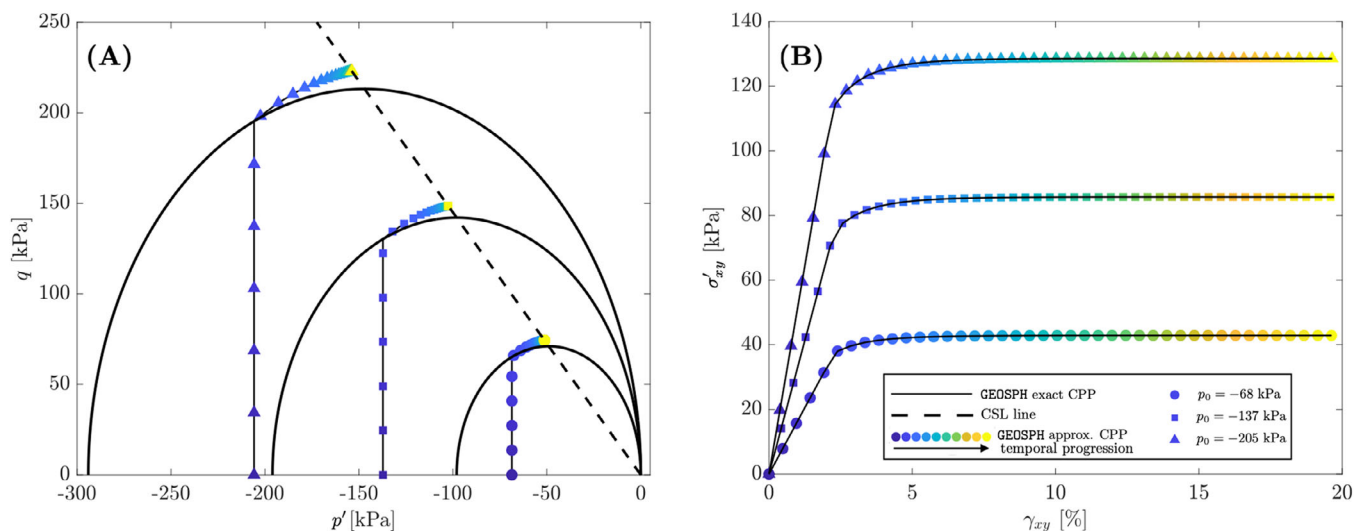


FIGURE B1 Comparison of the performance of approximate versus exact CPP algorithms for three different simple shear tests of overconsolidated soils. In the first test $p'_0 = -68$ kPa and $(p_c)_0 = -98$ kPa, in the second $p'_0 = -137$ kPa and $(p_c)_0 = -196$ kPa, and in the third $p'_0 = -205$ kPa and $(p_c)_0 = -294$ kPa.

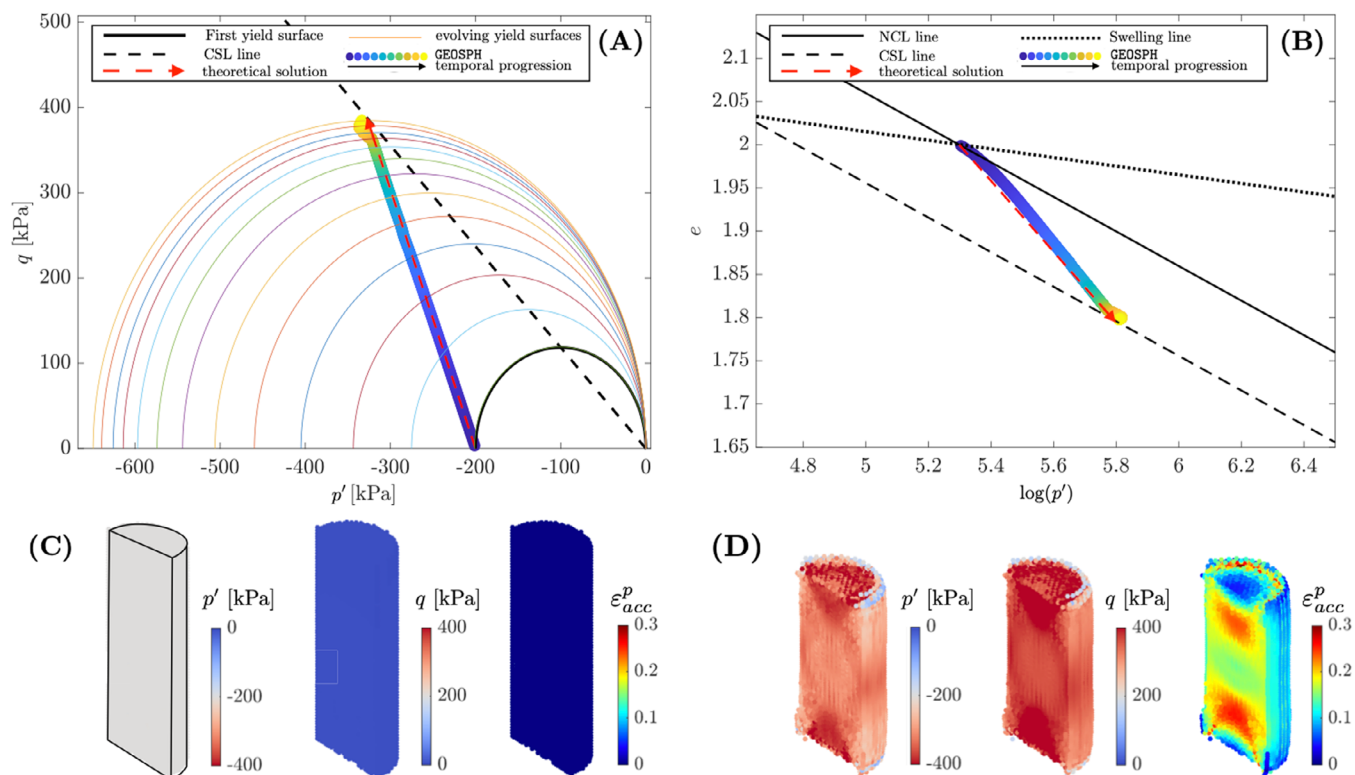


FIGURE B2 Same as Figure 5 but for the drained normally consolidated soil (TD-N) simulation. Outline of cylinder added for extra visibility of the $p' = -200$ kPa contour plot.

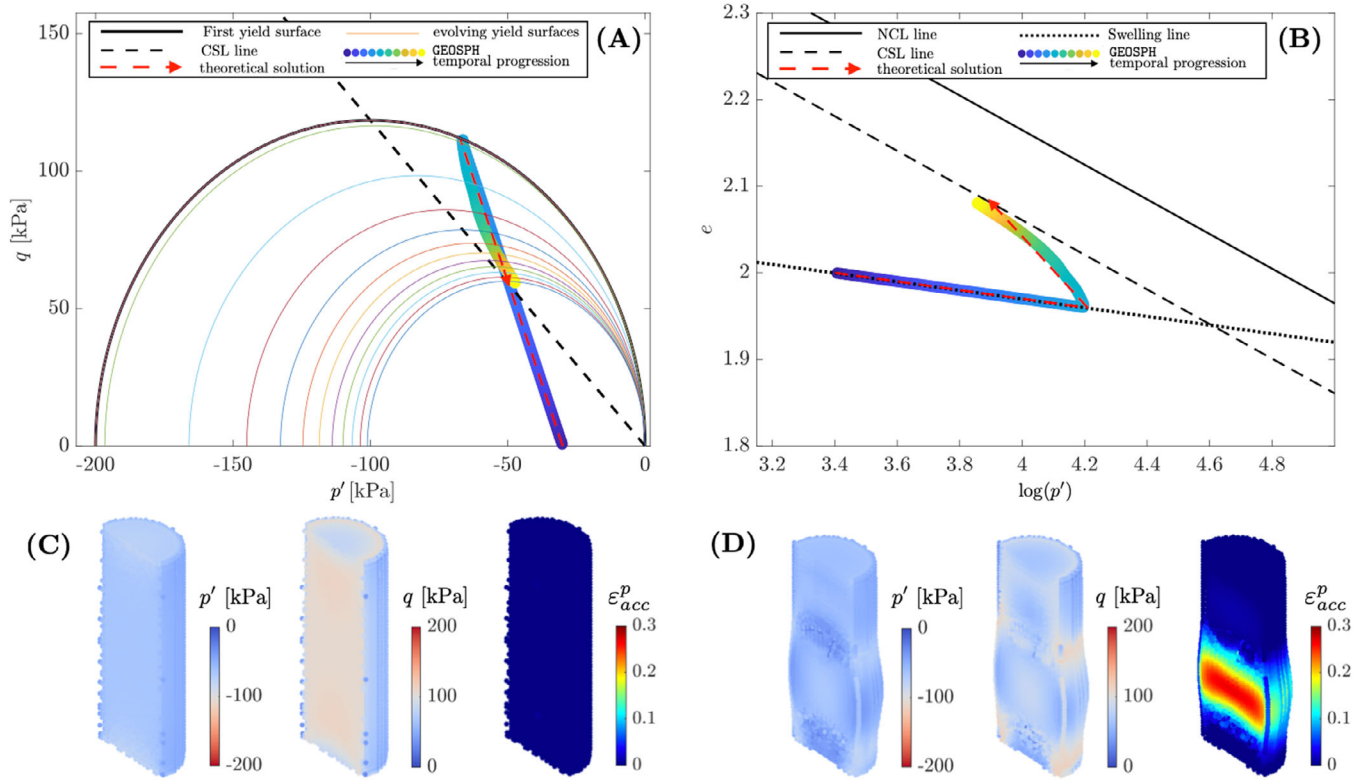


FIGURE B3 Same as Figure 5 but for the drained moderately overconsolidated soil (TD-M) simulation.

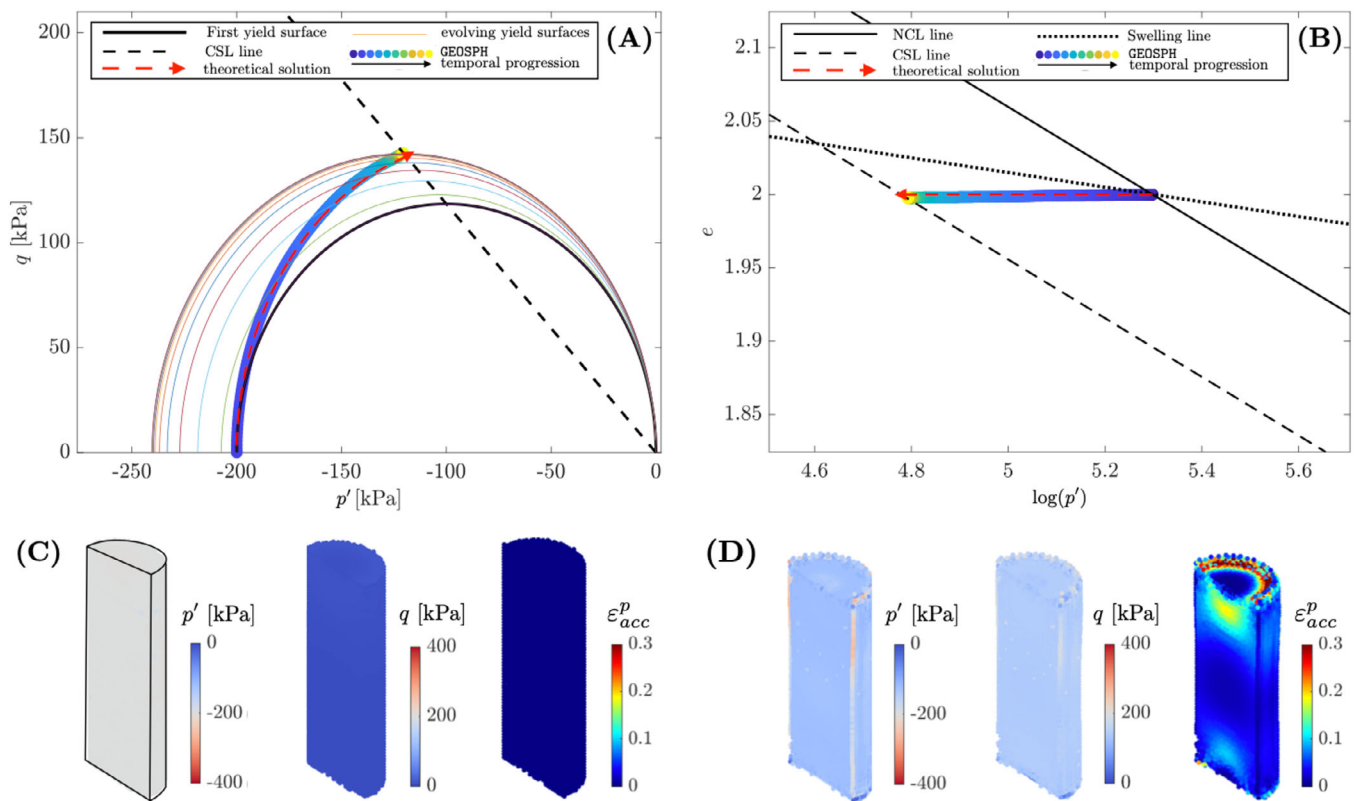


FIGURE B4 Same as Figure 5 but for the undrained normally consolidated soil (TU-N) simulation. Outline of cylinder added for extra visibility of the $p' = -200$ kPa contour plot.

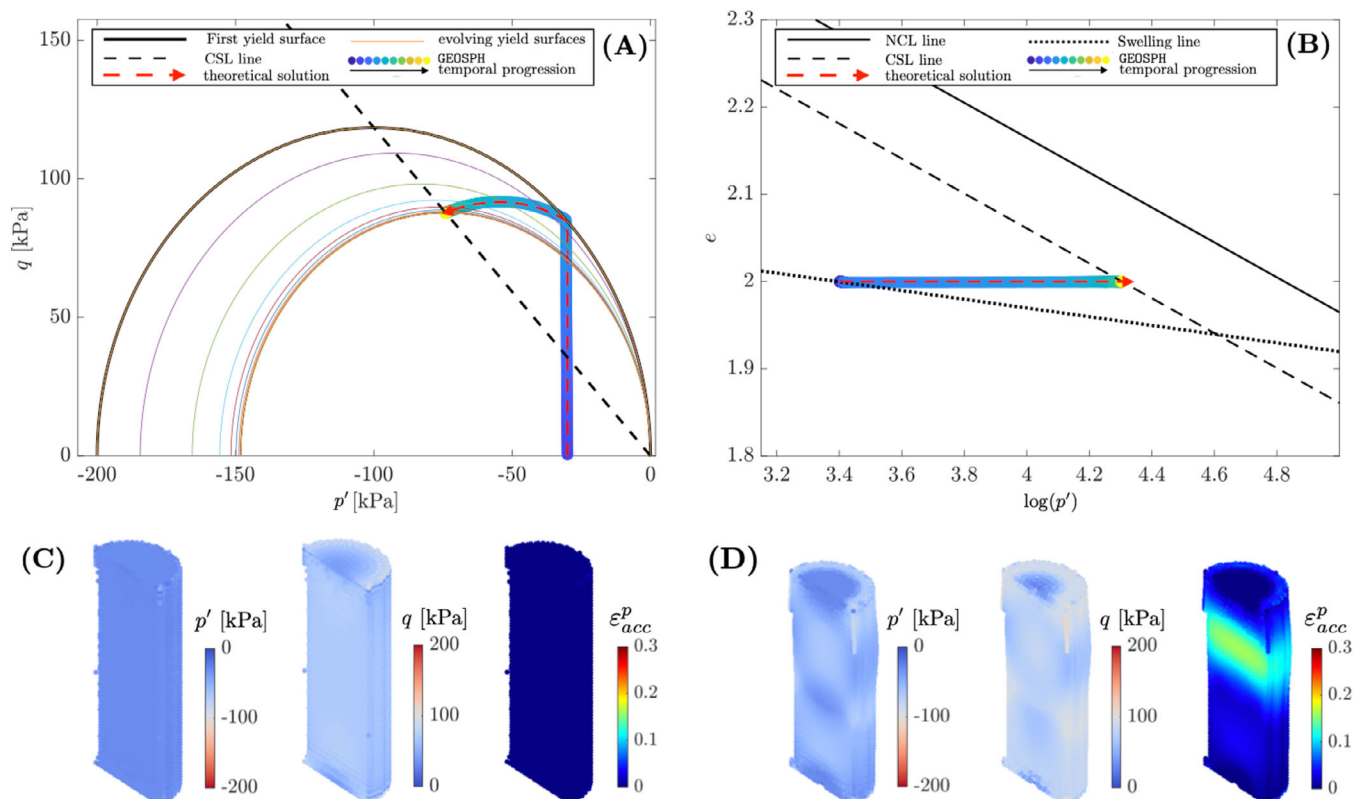


FIGURE B5 Same as Figure 5 but for the undrained moderately consolidated soil (TU-M) simulation.

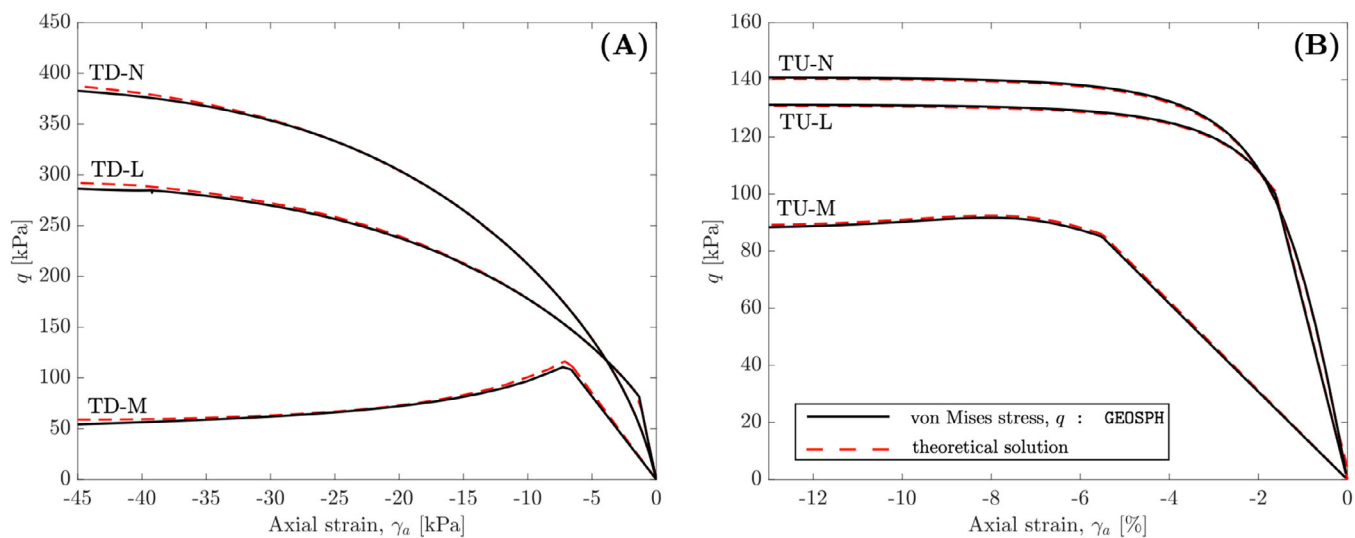


FIGURE B6 Evolution of the von Mises stress q from the three drained (A) and three undrained (B) triaxial test simulations compared to the theoretical von Mises stress.⁷⁵